TissuePMHC: A Matched-Pair Benchmark and Multi-Task Framework for Tissue-Specific MHC-I Presentation Preference

Anonymous Authors

Affiliations withheld for anonymous review.

***Abstract*—**MHC-I presentation can vary among tissues even when the presenting allele and source protein are fixed. Existing predictors largely estimate tissue-blind peptide–MHC compatibility and therefore do not directly address this tissue-associated preference. We formulate a matched-pair ranking problem: given two peptides from the same parent protein under the same presenting restriction, determine which has stronger presentation evidence in a specified tissue. The benchmark contains 157 human tissue–HLA combinations and 24 mouse tissue–H-2 combinations, where H-2 is the mouse major histocompatibility complex; each contains more than 200 matched peptide pairs. TissuePMHC combines a peptide representation learned across all human tasks with an allele-restricted representation, while species-specific baselines and dual-branch models are evaluated independently in mouse. On saved internal test sets, the strongest retained prediction aggregates have equal-weight mean AUROC/AUPRC values of 0.8448/0.8348 in human and 0.8528/0.8477 in mouse. When every held-out peptide identity is excluded from all training tasks, the highest identical-fold AUROC values are 0.7652 and 0.7590, respectively. Removing tissue identity substantially reduces human performance, showing that general peptide–MHC compatibility alone does not reproduce the complete ranking signal. The full human architecture is statistically comparable to its strongest simpler guided control, while mouse auxiliary supervision provides a small but consistent improvement over the matched unguided dual branch. These results support learnability of tissue-conditioned peptide ranking for previously represented tissue–MHC combinations and quantify the optimism associated with cross-task peptide reuse in human. They do not establish a causal tissue-processing mechanism, generalization to unseen tissues or alleles, independent-cohort validity, or clinical utility.

***Index Terms*—**Tissue-specific antigen presentation, MHC-I, matched-pair benchmark, multi-task learning, peptide presentation.

1. Introduction

MHC-I molecules display intracellularly derived peptides at the cell surface for surveillance by cytotoxic CD8^{+} T cells. The resulting immunopeptidome is shaped by substantially more than peptide binding to an MHC: protein abundance and turnover, proteolytic degradation, antigen-processing pathways, cellular composition, tissue environment, and experimental detectability can all affect which peptides are observed. Multi-tissue immunopeptidomic studies have consequently revealed both widely shared ligands and pronounced tissue-associated differences [2, 3]. Nevertheless, most computational models are designed to predict peptide binding to or presentation by an MHC in a general setting and therefore do not directly characterize how presentation evidence varies across tissues under a fixed MHC context.

Tissue-associated presentation preference by MHC-I can arise at several successive stages of antigen processing. Genes encoding constitutive or immunoproteasomal subunits, TAP1/2 transporters, ERAP1/2 aminopeptidases, and components of the peptide-loading machinery are not expressed uniformly across tissues. Together with tissue-dependent source-protein abundance and cell-type composition, these differences can alter which proteins are cleaved, which peptide fragments enter the endoplasmic reticulum, how precursor peptides are trimmed, and which sufficiently stable peptide–MHC-I complexes are ultimately displayed at the cell surface [4]. Thus, the same source protein may generate different presented epitopes in different tissues, while the same epitope may differ across tissues in the amount or likelihood of surface presentation. Understanding this variation is important for cancer therapy: it can help prioritize epitopes that are efficiently presented by tumors but minimally presented by essential normal tissues, thereby improving target selection and reducing potential on-target, off-tumor toxicity. In the present study, these mechanisms provide a biological explanation for tissue-associated preference rather than a causal claim, because the model does not directly measure gene expression or the activity of individual processing steps.

Existing tools answer related but different questions. Some estimate whether a peptide can bind an MHC; others estimate whether it is likely to be presented in a pooled or sample-specific setting, sometimes using expression or processing information [6, 7, 9]. Here we ask a narrower relative question: when the MHC allele and source protein are held constant, which of two peptides has stronger evidence in one specified tissue? Holding these two factors constant reduces easy explanations based only on peptide–MHC compatibility or protein identity and makes tissue-associated differences the focus of the comparison.

We call this question tissue–MHC-specific presentation preference prediction. Each prediction context corresponds to one tissue and one MHC restriction. Let p denote a peptide and let $\tau = (t, m)$ denote that tissue–MHC combination. A model produces a score

$$s_\theta(p, \tau) \in \mathbb{R},$$

which is used to rank peptides within that combination. A larger score means stronger predicted evidence relative to other candidates in the same context; it is not an absolute probability that the peptide is present. The model is evaluated only for tissue–MHC combinations represented during training, so it should not be interpreted as a predictor for a new tissue, a new MHC allele, or a new tissue–MHC combination.

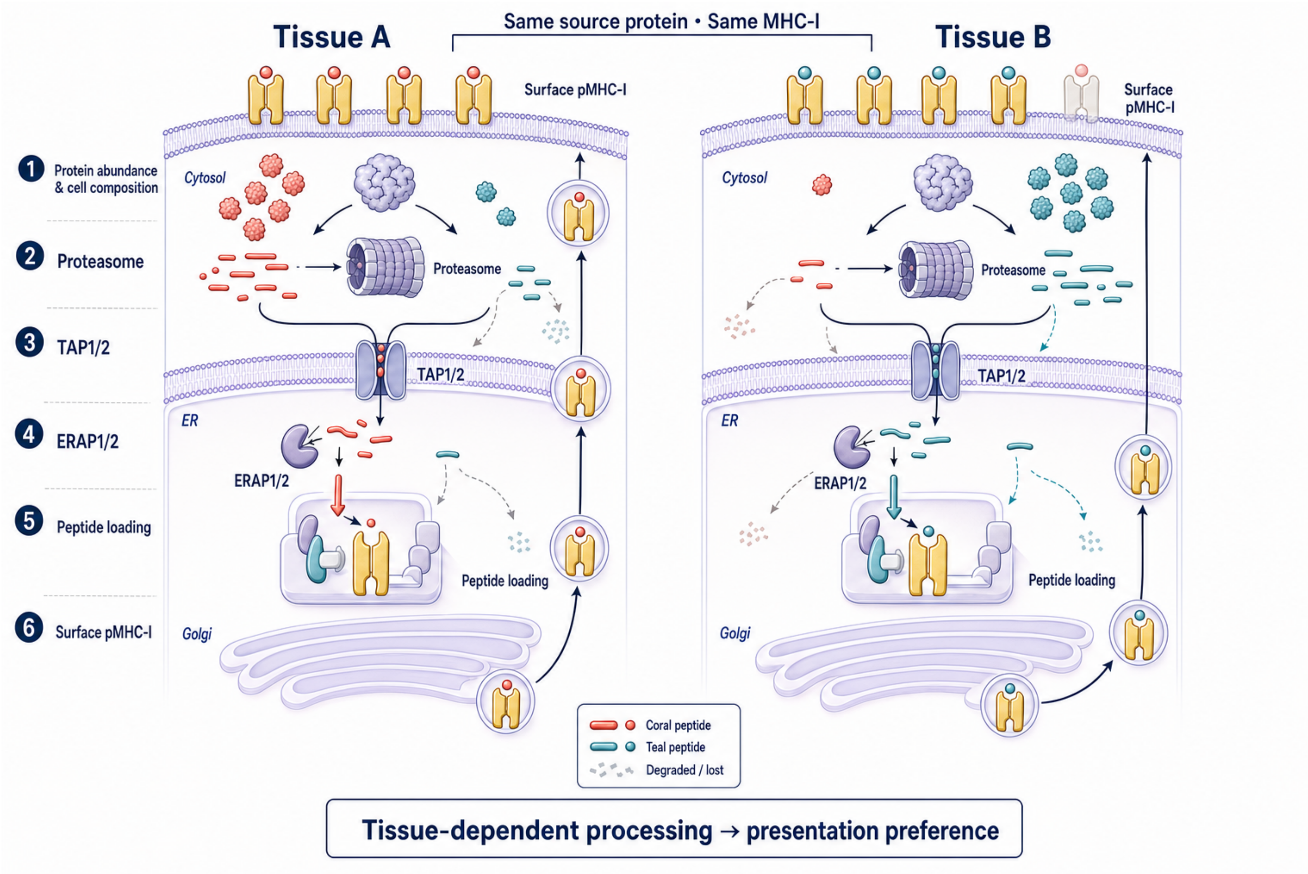

Figure 1: Conceptual mechanisms that can generate tissue-associated presentation preference by MHC-I under a fixed source-protein and MHC context. Tissue-dependent source-protein abundance and cell-type composition, TAP-dependent transport, ERAP-mediated trimming, and peptide loading can enrich different peptides for surface display. Dashed paths denote peptide degradation or loss from the presented repertoire, not confirmed biological absence. The schematic motivates an association and does not establish causality for any individual processing step.

To construct training examples aligned with this objective, each recorded-positive peptide is paired with a comparison peptide that has evidence elsewhere but not in the target tissue (a pseudo-negative peptide) under the constraints

$$m^+ = m^-, \qquad u^+ = u^-,$$

where m is the MHC restriction and u is the parent UniProt protein. One peptide is reported in the target tissue; its comparison peptide is reported elsewhere under the same MHC and from the same protein, but not in the target context. Biologically, the pair asks whether two fragments of one protein show different tissue-associated evidence after controlling the presenting allele and source-protein identity. The second peptide is a pseudo-negative rather than a confirmed biological absence. Missing reports can reflect incomplete sampling or limited mass-spectrometry detection, and the pairing cannot remove study, batch, expression, or cell-composition effects.

On this basis, we introduce paired human and mouse TissuePMHC benchmarks and a human prediction model. The model combines a globally shared peptide representation with a representation restricted to one HLA allele; in biological terms, this allows broadly shared sequence patterns and allele-associated motifs to contribute separately. In the ordinary internal evaluation, a held-out peptide is absent from fitting rows for the same tissue–MHC combination but may recur in another combination. We also perform a more demanding analysis in which every held-out peptide is removed from all fitting combinations. The latter tests tissue-associated ranking for unseen peptide identities within previously represented tissue–MHC contexts.

The study makes four main contributions. First, it defines a biologically motivated comparison in which MHC restriction and parent protein are matched. Second, it provides human and mouse benchmarks that use the same minimum number of pairs for retaining a tissue–MHC combination. Third, it measures how much performance falls when exact peptide identities cannot be reused across training and evaluation. Fourth, it tests whether the remaining signal depends on tissue identity or can be explained by general peptide–MHC compatibility alone. The aim is therefore not only to report a high score, but also to clarify what biological question each score can and cannot answer.

2. Related Work

2.1. Peptide–MHC Binding and Presentation Prediction

Peptide–MHC binding prediction is a foundational problem in computational immunology. Established methods such as

NetMHCpan [6], MHCflurry [7], and ACME [8] estimate binding affinity, binding rank, or peptide–MHC compatibility from peptide sequences and allele information. Pan-allelic predictors improve coverage by representing MHC molecules through pseudo-sequences that summarize the peptide-binding groove or through related allele features. This representation allows information sharing across well-characterized and data-scarce alleles [5, 6]. These models are essential for filtering candidate epitopes, but binding is only one stage of antigen presentation. A peptide with favorable binding characteristics may still be absent from the cell-surface immunopeptidome because its source protein is not expressed or degraded, the peptide is not generated or transported efficiently, or it is not detected experimentally. Binding scores therefore do not directly encode presentation-evidence preferences among tissues that share the same MHC restriction.

2.2. Predicting Cell-surface Antigen Presentation (Antigen Presentation Prediction)

Eluted-ligand data enable models to learn a broader presentation signal than binding measurements alone [6, 7]. General presentation predictors combine peptide sequence, MHC context, and, in some cases, antigen-processing or source-protein features to distinguish observed ligands from background peptides. Systems such as BigMHC [9] illustrate how large immunopeptidomic collections and modern sequence models can improve ligand presentation prediction and candidate ranking. Their usual target, however, is whether a peptide is likely to be presented for an allele, sample, or pooled presentation setting. This target differs from the relative question studied here: given a particular tissue–MHC context, which of two peptides from the same parent protein has stronger tissue-conditioned presentation evidence? Consequently, a high general-presentation score need not imply a tissue-specific preference, and direct comparisons require a benchmark whose supervision reflects that distinction.

2.3. Tissue- and Sample-Context Immunopeptidomics

Multi-tissue immunopeptidomic atlases have revealed that ligand repertoires contain both broadly shared peptides and tissue-associated components [2, 3]. MHCflurry 2.0 incorporates antigen-processing features but does not use tissue identity [7]. More recent systems perform multi-allelic deconvolution, as in ImmuneApp [20], or combine peptide, MHC, peptide-flanking sequence, and optional gene-expression features, as in HLApollo [21]. These approaches demonstrate that presentation is context dependent, but they generally do not define a tissue–MHC combination as the supervised prediction unit or match every recorded positive and pseudo-negative on both MHC restriction and parent protein. Our formulation is therefore complementary to expression- and sample-aware prediction: it isolates a ranking problem conditioned on a known biological context that can later be combined with molecular measurements.

2.4. Sharing Information across Tissue–MHC Combinations

Some tissue–MHC combinations contain many observations, whereas others contain relatively few. Learning a completely separate model for every combination would waste information shared across tissues and alleles. Conversely, pooling all combinations into one model could blur allele-specific binding motifs and tissue-associated differences. Joint learning provides a middle ground: related combinations share a peptide representation while retaining separate prediction functions [10, 11]. More structured methods allow different groups to use shared and specialized model components [12, 13]. TissuePMHC adopts a biologically structured sharing scheme: one model view learns patterns across all human combinations, while a second shares information only among tissues with the same HLA allele.

2.5. Relation to This Work

The principal distinction from previous work is the prediction target. General predictors estimate peptide binding to or presentation by an MHC, whereas TissuePMHC ranks peptides matched by MHC restriction and parent protein within a specified tissue–MHC context. The human and mouse datasets define this comparison identically, although each species uses a separately trained model. We distinguish ordinary internal evaluation, which permits a peptide to recur in another fitting combination, from peptide-separated evaluation, in which a held-out peptide is absent from every fitting combination. MHCflurry and NetMHCpan are therefore used as tissue-independent peptide–MHC controls rather than as tissue-aware competitors.

Table 1 makes these distinctions explicit. Negative construction varies across implementations and datasets; the entries therefore summarize the published prediction design rather than asserting identical data provenance.

Table 1 compares prediction designs rather than numerical performance.

Table 1: Comparison with representative methods for peptide binding or presentation. “Context” denotes information used by the published method beyond the peptide and MHC.

Method	Prediction target	Context	Negative/evaluation emphasis
NetMHCpan 4.1 [6]	Binding and presentation learned from eluted ligands	MHC pseudo-sequence; no tissue identity	Pan-allelic prediction
MHCflurry 2.0 [7]	Binding and presentation	Antigen-processing features; no tissue identity	Prediction across many alleles (pan-allelic general prediction)
BigMHC [9]	Presentation and immunogenicity	Peptide and MHC	Large-scale general prediction
ImmuneApp [20]	Epitope prediction and HLA-I deconvolution	Mono- and multi-allelic ligand data	General or sample-level evaluation
HLApollo [21]	Peptide presentation	MHC, peptide flanks, and optional gene expression	Alternative negative-set definitions; pan-allelic evaluation
TissuePMHC benchmark	Tissue–MHC presentation preference	Tissue and MHC identity	Peptide pairs matched by MHC and parent protein; ordinary and peptide-separated internal evaluation

3. Problem Formulation and Benchmark

3.1. Defining a Tissue–MHC Prediction Context

One prediction context corresponds to one tissue and one MHC-I restriction. For example, lung with HLA-A*02:01 and liver with the same allele are different contexts because the tissue differs. Let t denote a tissue and m an MHC-I restriction. We define the set of observed contexts as

$$\mathcal{T} = \{\tau = (t, m)\}.$$

Each benchmark row is represented by

$$x_i = (p_i, \tau_i), \qquad y_i \in \{0, 1\},$$

where p_i is a nine-residue peptide and τ_i identifies one tissue–MHC context. A positive label, $y_i = 1$, indicates positive ligand evidence in the target context. A comparison label, $y_i = 0$, indicates positive evidence under the same MHC restriction and parent protein in at least one other tissue, but no report in the target context. Thus, the pseudo-negative label represents absence of recorded evidence rather than experimentally verified non-presentation.

Given (p, τ), a model parameterized by θ produces

$$s_\theta(p, \tau) \in \mathbb{R},$$

with larger values indicating stronger predicted evidence within tissue–MHC combination τ. Operationally, the model learns to rank a peptide reported in the target tissue above a comparison peptide matched by MHC restriction and parent protein. It is trained on individual peptide rows, while the paired construction controls example generation and evaluation. Because each tissue–MHC combination has its own prediction function, the benchmark evaluates previously represented combinations; it does not assess unseen tissues or unseen MHC restrictions.

3.2. Pairing Peptides by Tissue, MHC Restriction, and Parent Protein

Each pair j contains one positive peptide p_j^+ and one pseudo-negative peptide p_j^- with evidence elsewhere but not in the target tissue. The two rows share the same target tissue, MHC restriction, and parent UniProt protein:

$$t_j^+ = t_j^-, \qquad m_j^+ = m_j^-, \qquad u_j^+ = u_j^-.$$

This matching removes two easy explanations for a difference: the peptides cannot be separated merely because they use different MHC molecules or come from different proteins. The remaining difference can include tissue-associated processing and presentation, but also differences in cell composition, study design, detection sensitivity, and annotation completeness. The benchmark therefore measures tissue-associated evidence, not a purified molecular mechanism.

We first calculate performance separately for every tissue–MHC combination and then give each combination equal weight, so common combinations do not dominate rare ones. We also report within-pair ordering accuracy:

$$\mathrm{Acc}_{\mathrm{pair}} = \frac{1}{N} \sum_{j=1}^{N} \mathbf{1}\left[s_\theta(p_j^+, \tau_j) > s_\theta(p_j^-, \tau_j) \right].$$

where N is the number of held-out pairs. This quantity is the fraction of pairs for which the peptide reported in the target tissue receives the higher score. AUROC measures ordering across all rows within a tissue–MHC combination, whereas within-pair accuracy directly evaluates the original MHC- and parent-matched comparison.

3.3. Evidence Inclusion Criteria

The human and mouse benchmarks are derived from the full ligand export of the IEDB downloaded on July 4, 2026 [1]. We apply the same biological inclusion rules to both species. We retain only clearly positive ligand records for MHC-I with an identifiable parent protein and a standard unmodified 9-amino-acid peptide:

- a qualitative measurement whose normalized text begins with *positive*;
- a MHC-I restriction, with both peptide source and host matching Homo sapiens for the human benchmark or Mus musculus for the mouse benchmark;
- a human restriction matching the four-digit HLA pattern `HLA-...*dd:dd` (the extractor also accepts the equivalent colon-free four-digit form) or a restriction from the mouse major histocompatibility complex (H-2; written as `H2` in the processed identifiers) beginning with `H2-`;
- a parent UniProt accession extracted from the molecule-parent web identifier (internationalized resource identifier; IRI) in the IEDB;
- a linear, unmodified peptide containing only the 20 standard amino acids;
- a peptide length of exactly nine residues; and
- deduplication before pairing by the exact peptide, restriction, tissue, and parent/source-protein annotation fields.

Source-tissue values are stripped of surrounding whitespace and retained as the exact names used in the IEDB; distinct nonempty tissue strings are not merged after extraction. Empty or missing-value-like fields required to define a tissue–MHC combination are removed before benchmark filtering, excluding 857 otherwise eligible human pairs. Pair-consistency checks require both rows to have the same nonempty tissue and restriction. Retained HLA and H-2 strings are used directly as identifiers after the pattern checks above. Records without a parent UniProt accession are excluded because parent-protein matching is a defining control of the benchmark. Restricting the study to unmodified canonical 9-residue peptides for MHC-I provides a consistent sequence space, but excludes variable-length ligands, post-translationally modified peptides, and presentation by MHC-II.

3.4. Constructing MHC- and Parent-matched Peptide Pairs

For each target tissue–MHC combination, we compare a peptide reported in that tissue with another peptide from the same protein that is reported in a different tissue under the same MHC restriction. A fixed random state, 20260704, makes the following construction reproducible:

1. Group all eligible ligand records by MHC restriction m and parent UniProt protein u.
2. For each target combination (t, m, u), collect peptides with positive evidence in tissue t.
3. From other tissues represented in the same (m, u) group, collect candidate comparison peptides with evidence elsewhere but not yet in the target tissue (pseudo-negative peptides).
4. Exclude every candidate with a positive report in the target tissue–MHC combination (t, m).
5. Within each target combination and label, use each peptide at most once. For a given (t, m, u) group, set the number of pairs to the smaller of the eligible positive and pseudo-negative counts and sample both sets without replacement.
6. Randomly permute the selected pseudo-negatives and join them one-to-one with the selected positives.
7. Assign a persistent pair identifier, preserve the reported-tissue and protein annotations, and verify that every identifier has exactly one row per label, identical tissue, restriction, and parent UniProt values in both rows, and no target-tissue report for the pseudo-negative.

Table 2: Human benchmark and saved internal test-set statistics.

Property	Value
Tissue–HLA combinations	157
Tissues	25
HLA alleles	35
Minimum eligible pairs per retained combination	201
Training pairs	73,798
Test pairs	15,700
Training rows	147,596
Test rows	31,400
Test pairs per combination	100

Every pair contributes one target-tissue peptide and one comparison peptide, so the dataset is deliberately balanced. This balance is useful for controlled ranking but does not resemble the natural frequency of presented peptides in a tissue sample. The reported scores therefore describe relative ranking within this constructed benchmark, not the probability that a peptide would be detected in a biological specimen.

3.5. Human Benchmark

The human and mouse benchmarks use the same eligibility rule: a tissue–MHC combination is retained only when it contains more than 200 eligible MHC- and parent-matched peptide pairs. Under this rule, the human benchmark contains 157 tissue–HLA combinations, of which the smallest contains 201 pairs. Within each retained combination, 100 complete pairs are sampled without replacement for a saved internal test set, and all remaining pairs form the training partition. Random state 20260704 makes this division reproducible.

The 157 tissue–HLA combinations summarized in Table 2 form an incomplete and long-tailed tissue-by-HLA coverage matrix. Performance is therefore calculated separately for each combination before averaging with equal weight; pooling all rows would overemphasize combinations with more examples and could conceal weak performance in sparsely represented contexts. The saved internal test set excludes complete pair identifiers, but a peptide or parent protein may still occur in a different combination on both sides of the split. It therefore evaluates familiar tissue–HLA contexts

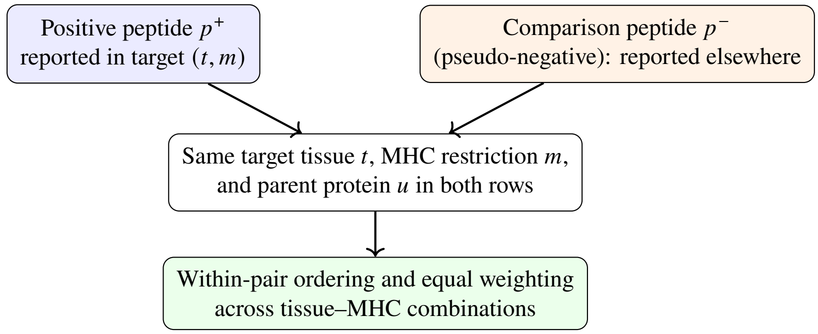

Figure 2: Construction of a peptide pair matched by target tissue, MHC restriction, and parent protein. The comparison peptide has positive evidence elsewhere but no report in the target tissue–MHC context.

Table 3: Mouse benchmark and saved internal test-set statistics.

Property	Value
Tissue–H2 combinations	24
Tissues	13
H-2 restrictions	4
Minimum eligible pairs per retained combination	224
Training pairs	6,766
Saved test pairs	2,400
Training rows	13,532
Saved test rows	4,800

rather than biological entities that are entirely absent from training.

For peptide-separated cross-validation, the complete 73,798-pair human training pool is considered jointly, while the separate 15,700-pair test set remains excluded. Pairs sharing either peptide are joined by following direct and indirect links before folds are formed. This produces 17,036 groups, including 10,923 containing multiple pairs; the largest contains 755 pairs. A reproducible assignment distributes complete groups across three folds while balancing the tissue–HLA combinations. Each held-out fold contains 24,598–24,600 pairs, each fitting set contains 49,198–49,200 pairs, all 157 combinations remain represented, and neither pair identifiers nor exact peptide sequences overlap between fitting and held-out data.

3.6. Mouse Benchmark

The mouse benchmark uses the same tissue-conditioned definition: the two peptides share their H2 restriction and parent protein, and a tissue–H2 combination is retained only when it contains more than 200 eligible pairs. This rule retains 24 tissue–H2 combinations; the smallest contains 224 pairs. Exactly 100 complete matched pairs per retained combination are sampled without replacement for the saved internal test set using random state 20260704, and the remainder form the training partition.

The mouse inventory is summarized in Table 3. This saved internal partition was not examined until the model and analysis choices had been specified. A held-out peptide is absent from the training rows of the tissue–H2 combination being evaluated, but it may occur in training rows for another combination. Table 4 makes this distinction explicit. For each test row, the audit first asks whether the same entity occurs in training for the current tissue–H2 combination and, only if it does not, whether it occurs in another combination. The three columns are therefore mutually exclusive.

Thus, a test peptide is new to the specific tissue–H2 combination being evaluated, even though most test rows contain a peptide already observed while fitting another combination. Among unique entities, 2,453 of 3,011 test peptides (81.47%) and 967 of 1,088 parent proteins (88.88%) occur somewhere in training. Parent proteins can also recur within the current combination because several matched pairs may come from the same protein. All 24 tissue–H2 combinations are represented in training. The saved test set is therefore an internal confirmation in familiar contexts with peptide reuse across contexts; it is not an evaluation of peptides, proteins, or biological combinations absent from all training, nor an independent external validation.

Mouse peptide-separated cross-validation applies the same transitive grouping of peptide-linked pairs to all 6,766 training-pool pairs. Complete groups are assigned to three folds while retaining all 24 tissue–H2 combinations in every held-out partition. Pair-identifier and exact-peptide overlap are both zero; the largest group contains 50 pairs, and each combination contributes 41–157 held-out pairs and 82–314 fitting pairs. These counts show that exact peptide separation across all combinations is feasible without deleting overlapping pairs after an initial random split.

3.7. Evaluation Settings and Their Biological Meaning

The evaluation design has three distinct layers: how observations are divided between fitting and evaluation, how independently trained models are combined, and how saved predictions are compared statistically. These layers do not define separate biological experiments. Table 5 distinguishes them in reader-facing language. In cross-validation, the training pool is divided into three parts and every part is predicted once by a model fitted on the other two. Independent training runs start the same model from different random states, whereas the random state used to reproduce pair sampling or fold assignment does not create another trained model. Model structure, tuning choices, training runs, and prediction averaging are specified before the relevant evaluation results are inspected.

What is actually predicted. The benchmark is constructed from matched peptide pairs, but the main neural models

Table 4: Mouse entity-overlap audit for the saved internal test set. Counts are test-row occurrences, except for pair identifiers, which are counted once per pair. "Current combination" means the tissue–H2 combination of the test row.

Held-out entity (counted unit)	Current-combination training	Other-combination training only	Absent from all training
Pair identifier (2,400 pairs)	0 (0%)	0 (0%)	2,400 (100%)
Peptide identity (4,800 rows)	0 (0%)	4,047 (84.31%)	753 (15.69%)
Parent UniProt (4,800 rows)	710 (14.79%)	3,756 (78.25%)	334 (6.96%)

Table 5: Evaluation settings and their relationships. A data setting changes which observations are available for fitting and evaluation. A model procedure changes how models are trained or combined without defining a new data split. A statistical comparison operates only on saved predictions.

Reader-facing setting	Layer	What is done	Purpose and relationship to the other settings
Saved internal test set	Data setting	For each tissue–MHC combination, 100 complete positive–pseudo-negative pairs are kept outside the training file. A model fitted on all remaining rows scores this saved set once.	Main internal benchmark for familiar tissue–MHC contexts. It contains different rows from both cross-validation analyses and is not an external cohort.
Ordinary cross-validation with complete pairs kept together	Data setting	The training pool is divided into three folds within each tissue–MHC combination. Both rows from one pair always remain together; each row is predicted by a model fitted on the other two folds.	Cross-validation used for model development and internal checking. The same peptide may still occur in fitting rows for another tissue–MHC combination.
Ordinary cross-validation on the same pooled rows	Comparison	Complete pairs are kept together while exactly the same training-pool rows used by peptide-separated cross-validation are evaluated. Human results average three independent training runs and mouse results average five.	Ordinary reference for the peptide-separated analysis. Both analyses score the same rows, but their fold memberships are not identical or matched pair by pair.
Repeated training and prediction averaging on the saved test set	Model descriptor plus saved test	A previously chosen model is refitted on the full training file from each prespecified initialization; predictions on the saved test set are then averaged. The main mouse result uses five independent runs.	Prespecification and prediction averaging describe model estimation, not a new type of data split.

do not receive two peptides simultaneously. They score one peptide row at a time using its sequence and tissue–MHC identity. Pair identifiers are used to create balanced recorded-positive and pseudo-negative examples and to keep the two rows of a pair in the same data partition. Training applies a separate present/absent loss to each row, and AUROC and AUPRC compare all positive and pseudo-negative rows within a tissue–MHC combination. Within-pair ordering accuracy asks the complementary question of whether the recorded-positive peptide receives the higher score. Keeping both members of a pair together protects data integrity; it does not turn the main training objective into a direct two-peptide ranking loss.

Saved internal test set. For each tissue–MHC combination, the dataset builder randomly selects 100 complete pairs for the test file and assigns the remaining pairs to the training file. Because a peptide is used at most once within a combination during pair construction, a test peptide is absent from training rows for that same combination. The sequence may nevertheless occur in another fitting combination and can then influence shared model parameters. This test set therefore measures internal discrimination in already represented biological contexts while allowing information sharing across combinations. Saving the partition and analysis choices before evaluation does not imply that the test set has never influenced the wider research process. The human test set was examined repeatedly during long-term model development; the mouse test set was opened only after the model, training runs, and prediction-averaging rule had been specified.

Ordinary cross-validation with complete pairs kept together. This analysis uses only the training file. Within each tissue–MHC combination, complete pairs are distributed across three nearly equal folds. Two folds fit the model and the third is held out; rotating this role gives one prediction for every row in the pool. The model predicting a row has therefore fitted neither member of its pair. The construction rule also prevents exact peptide reuse within that tissue–MHC combination, but the peptide may still appear in a different fitting combination. This is a development-stage

Table 5: Guide to evaluation terms used in this study (continued).

Term used in the project	Layer	What is done	Purpose and relationship to other terms
Peptide-separated cross-validation	Data setting	Pairs linked by any shared peptide sequence are placed in the same fold. A held-out peptide sequence is therefore absent from fitting rows for every tissue–MHC combination.	Robustness test for new exact peptide identities in familiar tissue–MHC contexts. It does not require held-out tissues, MHC alleles, or parent proteins to be new.
Prediction averaging under peptide-separated cross-validation	Model procedure plus peptide-separated cross-validation	The same peptide-separated folds are used for every independent training run, and the held-out predictions are averaged across runs.	Repeated-model implementation of the preceding data setting; it is not an additional split.
Transitive grouping of peptide-linked pairs	Split algorithm	Shared-peptide links are followed transitively: if pair A shares a peptide with B and B shares one with C, all three enter the same fold.	The fold-building algorithm that enforces exact-peptide separation across all tissue–MHC combinations. It is not a separate biological setting.
Within-combination comparison of ordinary and peptide-separated results	Statistical analysis	For each tissue–MHC combination, the ordinary result is compared with the peptide-separated result, and the resulting differences are summarized across combinations.	Quantifies how sensitive performance is to the change in fold construction. Exact-peptide separation is not fully independent validation.

estimate for familiar biological contexts, not an independent validation dataset. It also differs from the saved test set in both the evaluated rows and the amount of fitting data, so the two numerical results can only be compared descriptively.

Ordinary cross-validation on the same pooled rows. For the direct sensitivity analysis, ordinary cross-validation is repeated on exactly the same training-pool rows as the peptide-separated analysis. The model family, independent training runs, number of folds, and pooled evaluation rows are the same, and fold sizes are closely matched within each tissue–MHC combination. Fold membership is nevertheless constructed independently: corresponding folds are not forced to contain the same pairs, proteins, or peptide-linked groups. The comparison is informative because both analyses calculate each metric from the same labeled rows for a given biological combination. It measures sensitivity to alternative training-data partitions rather than isolating a causal contribution from repeated peptide identities.

Cross-validation with held-out peptides absent from all fitting combinations. To remove exact peptide reuse across combinations, each peptide pair is represented as a point in a network. Two points are linked when either pair contains the same peptide sequence. Pairs connected directly or through intermediate links form one group, and that complete group is assigned to one fold. Consequently, neither member of a held-out pair nor any exact copy of those sequences in another combination can occur in the fitting rows. All tissue–MHC combinations remain represented during fitting. The analysis therefore asks whether the model can recognize tissue-associated sequence preferences for peptide identities it has not previously seen while operating in familiar tissue and MHC contexts.

This construction changes more than the presence of an exact duplicate. A shared peptide can connect a chain of otherwise different pairs, causing the entire chain to move together. The resulting folds can also differ in parent-protein composition and in the distribution of related sequence patterns. Table 6 illustrates this point: although parent protein is not an input to the main models and is not explicitly separated, its overlap decreases markedly when peptide-linked groups are kept intact. The performance difference is therefore best interpreted as sensitivity to separating exact peptide identities. It does not isolate a pure or mechanism-specific contribution from repeated peptides.

Prespecified models, cohort-relative score combination, and statistical comparison. The human model structure and the mouse comparison families, training budgets, independent initializations, and prediction-averaging rules were specified before the peptide-separated results were examined. This prevents direct tuning to those results, but it does not make the experiment fully nested: the model structures had already been developed using ordinary cross-validation predictions from the same training-pool rows. The peptide-separated analysis is therefore a robustness evaluation of previously developed model families. In addition, the human model converts the two pathway outputs to relative order positions among peptides in each held-out tissue–MHC combination before averaging them. This transformation uses no held-out labels, but the final score is relative to the other peptides evaluated in that cohort rather than a stand-alone calibrated probability. Mouse neural controls instead average probabilities across independent training runs.

The paired statistics are calculated after prediction and do not create another split. Ordinary and peptide-separated results are compared separately within each tissue–MHC combination because both analyses cover the same pooled evaluation rows. This correspondence reduces variation caused by evaluating different sets of biological combinations, but it does not remove variation arising from fold membership,

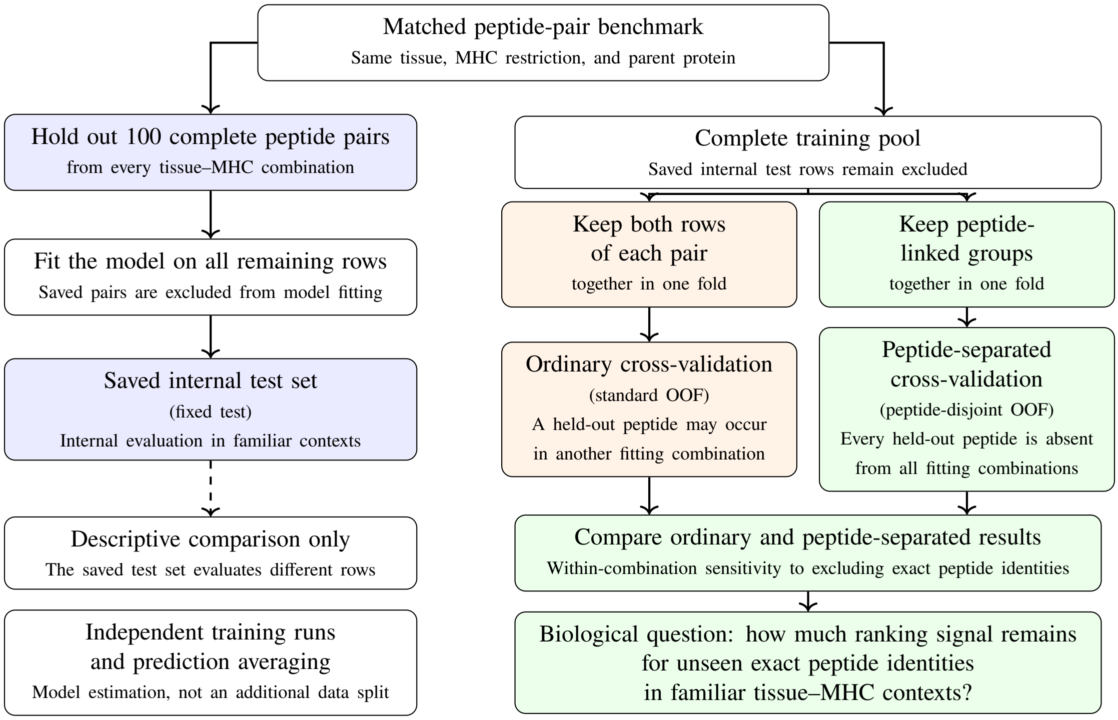

Figure 3: Relationship among the evaluation settings. The saved-test pathway evaluates different rows after fitting on the complete remaining training set. The two cross-validation pathways evaluate the same pooled rows but differ in what is kept together when folds are formed: complete pairs in the ordinary analysis, or entire groups connected by a shared peptide in the peptide-separated analysis. Repeated training runs and prediction averaging describe model estimation rather than a new data split. Only the two cross-validation results support a within-combination sensitivity comparison.

Table 6: Entity overlap in the two cross-validation analyses using the same pooled rows. Pair overlap is the number of complete pair identifiers shared between fitting and held-out rows. Peptide values give the fold range for the percentage of unique held-out peptide sequences also present in fitting. Protein values give the three-fold mean percentage of unique held-out parent proteins present in fitting.

Species	Cross-validation setting	Pair overlap	Peptide overlap	Protein overlap
Human	Ordinary cross-validation	0	69.19–69.91%	90.64%
Human	Peptide-separated cross-validation	0	0%	65.26%
Mouse	Ordinary cross-validation	0	71.43–72.12%	80.69%
Mouse	Peptide-separated cross-validation	0	0%	14.37%

peptide-linked group structure, or the resulting composition of the fitting data.

Scope of the biological conclusion. The ordinary evaluation asks whether peptide sequence helps distinguish recorded positives from matched pseudo-negatives in familiar tissue–MHC contexts while information may be shared across combinations. The peptide-separated evaluation asks whether a substantial part of that signal remains when the exact held-out sequences are removed from all fitting combinations. Neither analysis demonstrates that a pseudo-negative is biologically never presented, identifies a causal antigen-processing mechanism, or evaluates a new tissue, MHC allele, tissue–MHC combination, parent protein, study, or external cohort. Parent proteins, studies, experimental platforms, and related peptide motifs may still recur. The peptide-separated result is therefore an internal robustness analysis rather than fully independent biological validation.

4. Method: TissuePMHC Model for Human Data

4.1. Model Overview and Tissue–HLA-specific Prediction

Peptide presentation contains sequence patterns at more than one biological level. Some features may recur across many tissues and HLA alleles, whereas others may reflect motifs associated with one allele. TissuePMHC therefore contains a global pathway that learns from all tissue–HLA combinations with additional tissue and allele supervision, and an HLA-specific pathway that learns only from combinations sharing one HLA allele. Each pathway produces a separate score for every tissue–HLA combination. Their within-combination rankings are combined with equal weight and then averaged across three independently trained models.

Both pathways use a position-aware peptide encoder that preserves residue position and examines neighboring residues, followed by a separate output layer for each observed tissue–HLA combination. This structure allows combinations with many examples to support those with fewer

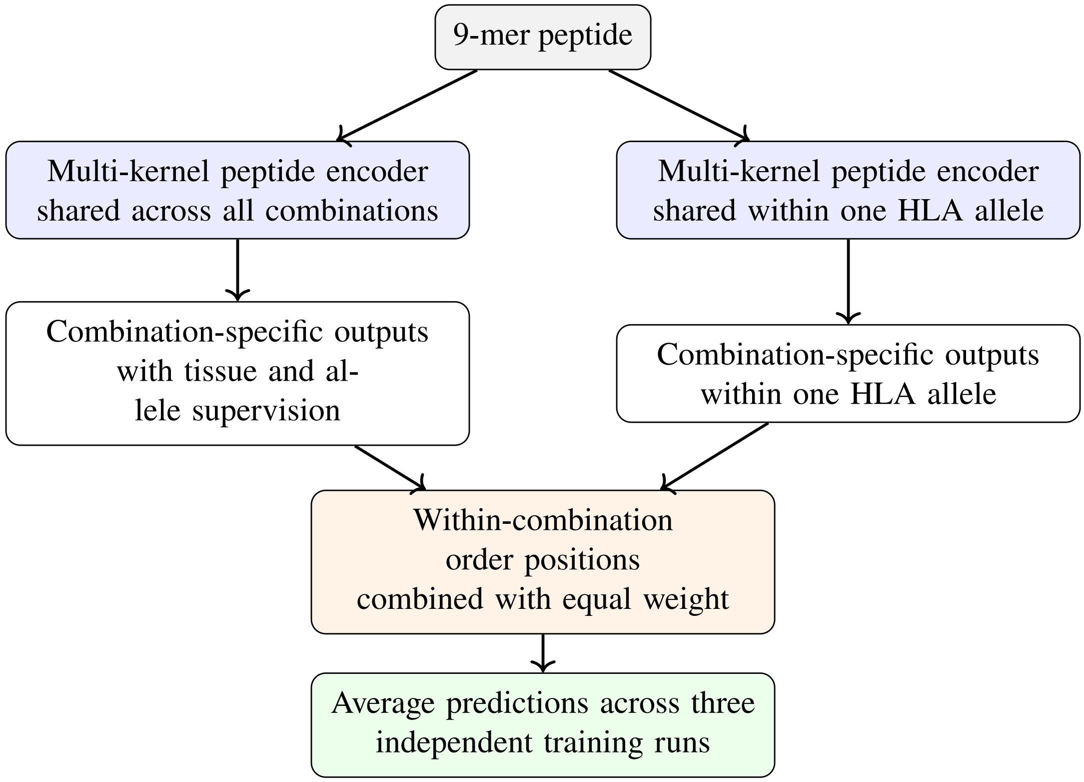

Figure 4: Overview of the human model. The global pathway learns peptide patterns across all tissue–HLA combinations, whereas the HLA-specific pathway shares information only among tissues with the same allele. Tissue and allele prediction provide training supervision but do not enter the final score. The two within-combination rankings are combined and averaged across three independent training runs.

examples without forcing all alleles to use an identical peptide representation. Because the final prediction depends on an output layer learned for a particular combination, the model is not designed for tissue–HLA combinations absent from training.

The remainder of this section gives the exact implementation needed for reproduction. Let h_i denote the numerical representation of peptide p_i, and let τ_i denote its tissue–HLA combination. Each pathway maintains a separate linear output for every available combination. The corresponding logit is

$$z_i = g_{\tau_i}(h_i),$$

and the associated presentation-preference probability is

$$q_i = \sigma(z_i),$$

where σ is the logistic sigmoid. During each mini-batch update, examples are routed to the output layer for their tissue–HLA combination, while encoder parameters are shared according to the pathway definition. The main objective is binary cross-entropy with logits, denoted BCE:

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{B}\sum_{i=1}^{B}\left[y_i \log q_i + (1-y_i)\log(1-q_i)\right].$$

Although every benchmark pair contains one recorded-positive row and one pseudo-negative row, the model processes rows independently. Pair identifiers are not used during the forward pass or loss calculation.

4.2. Peptide Encoding That Preserves Position and Examines Several Motif Lengths

The biological meaning of an amino acid depends on its position in a 9-mer peptide: anchor residues involved in MHC binding are not interchangeable with residues at other positions. The encoder therefore preserves all nine positions while examining local patterns of several lengths. For a peptide $p = (a_1, \ldots, a_9)$, each residue is mapped to a learned 16-dimensional embedding and combined with a learned positional embedding:

$$X = E(p) + P \in \mathbb{R}^{9\times d},$$

where $E(p)$ contains the residue embeddings and P is a trainable $9 \times d$ positional representation shared across peptides. These positional embeddings distinguish the same amino acid at different locations, which is important because anchor and supporting positions in MHC-I ligands are not interchangeable.

The model operates only on canonical 9-mer peptides. No tissue transcriptomics, proteomics, parent-protein sequence, flanking sequence, or external binding score is supplied as input. Tissue and HLA information enter through the combination identifiers, combination-specific output layers, parameter-sharing structure, and auxiliary training labels rather than through additional omics measurements.

To examine local sequence patterns spanning two, three, or five residues, we apply one-dimensional convolutions with widths

$$\mathcal{K} = \{2, 3, 5\}.$$

For each $k \in \mathcal{K}$, the input is transposed to a channels-first representation and processed as

$$H_k = \mathrm{ReLU}\left(\mathrm{Conv1D}_k(X)\right).$$

Each convolution produces 32 feature maps. Padding retains local context at the peptide ends. For even kernel widths, symmetric padding produces one additional position; the final position is therefore cropped to retain a length of nine. Each pathway consequently yields

$$H_k \in \mathbb{R}^{9\times 32}.$$

The three sets of local patterns are joined while preserving their sequence positions. We deliberately avoid reducing the peptide to a position-free maximum or average, because the same short motif can have a different meaning at a different position:

$$v = \mathrm{Flatten}\left[H_2; H_3; H_5\right] \in \mathbb{R}^{9\cdot 32\cdot 3}.$$

The resulting vector is passed through a two-layer multilayer perceptron,

$$h = \mathrm{MLP}(v) \in \mathbb{R}^{128},$$

Each hidden layer has 128 units with rectified linear unit activation. During training, dropout with probability 0.2 is applied after the first hidden layer. Flattening the position-specific feature maps allows identical local patterns at different peptide positions to contribute differently to the representation. The multi-kernel encoder is tailored to peptides of this single length; convolution itself is not presented as a methodological novelty.

4.3. Information Sharing Across All Combinations and Within HLA Alleles

The global pathway allows every tissue–HLA combination to contribute to a shared peptide representation. During training, auxiliary tissue and HLA classification objectives encourage this representation to retain biologically relevant grouping information. These objectives do not provide gene expression, protein abundance, or other molecular measurements. If $c_i^{(t)}$ and $c_i^{(m)}$ denote the tissue and HLA categories, respectively, the global objective combines BCE with categorical cross-entropy, denoted CE:

$$\mathcal{L}_{\mathrm{global}} = \mathcal{L}_{\mathrm{BCE}} + \lambda_t \mathcal{L}_{\mathrm{CE}}\left(\hat{c}^{(t)}, c^{(t)}\right) + \lambda_m \mathcal{L}_{\mathrm{CE}}\left(\hat{c}^{(m)}, c^{(m)}\right),$$

with prespecified weights

$$\lambda_t = \lambda_m = 0.1.$$

The tissue and HLA predictions guide representation learning but do not enter the final peptide score. They provide auxiliary supervision rather than independent biological evidence.

The HLA-specific pathway shares information only among tissues with the same allele, preserving allele-associated peptide preferences that broad pooling could blur. For each HLA allele m, we define

$$\mathcal{T}_m = \{(t, m) \in \mathcal{T}\}.$$

A separate encoder is trained on rows belonging to $\mathcal{T}_m$, and each tissue–HLA combination in the group receives its own linear output layer. The objective for allele m is

$$\mathcal{L}_m = \frac{1}{|\mathcal{D}_m|} \sum_{i\in\mathcal{D}_m} \mathcal{L}_{\mathrm{BCE}}\left(g_{\tau_i}^{(m)}(h_i^{(m)}), y_i\right).$$

This pathway has no auxiliary tissue or HLA classification objective. Because every example handled by one allele-restricted model shares the same allele, the pathway concentrates on allele-specific peptide patterns and variation among tissues under that restriction. It complements the global pathway: broad sharing can stabilize combinations with few examples, whereas allele-restricted sharing can preserve motif structure that unrestricted pooling may blur.

4.4. Combining the Two Within-combination Orderings

The two pathways are trained separately, so a score of 0.8 need not have the same calibration in both. We therefore combine their rankings rather than their raw scores. Within each tissue–HLA combination, every candidate is converted to a percentile rank. For combination τ, let $q_{g,i}$ and $q_{h,i}$ be the two scores for row i:

$$r_{g,i} = \mathrm{PercentileRank}_\tau(q_{g,i}),$$
$$r_{h,i} = \mathrm{PercentileRank}_\tau(q_{h,i}).$$

The equal-weight rank-fused prediction for one training run is

$$s_i^{(b)} = \frac{r_{g,i} + r_{h,i}}{2}.$$

Both pathways receive equal weight; no tissue or test set is used to choose a preferred pathway. This procedure answers a ranking question, not an absolute probability question. A peptide's percentile depends on the other candidates scored in the same batch, so adding or removing candidates can change its final rank. Prospective use would therefore require either a predefined candidate set or a prespecified reference distribution.

4.5. Averaging Independent Training Runs and Its Scope

Neural-network training can vary across random initializations. We therefore repeat the complete training procedure three times and average the resulting rankings [15]:

$$s_{i,\mathrm{final}} = \frac{1}{B}\sum_{b=1}^{B} s_i^{(b)}, \qquad B = 3.$$

For each training run, the probabilities from both pathways are first converted to within-combination ranks and combined. The resulting scores are then averaged across runs. This aggregation order is used consistently for the saved internal test set, pair-preserving cross-validation, and cross-validation in which each held-out peptide is excluded from every training combination.

The averaged prediction is a single point estimate and should not be confused with training stability. We therefore report results from the individual runs as well as their prediction average, separating variation across initializations from the benefit of averaging.

4.6. Species-specific Mouse Models

The mouse analysis tests whether the same prediction problem is learnable in a second species; no human parameters are transferred to mouse. Mouse implementations are trained independently on the 24 tissue–H2 combinations and include simple sequence baselines, a shared encoder, and matched dual-branch models with and without auxiliary tissue and H2 supervision. These models follow the control definitions in the Experimental Setup and are compared on identical folds. Neural predictions in the peptide-separated analysis are averaged across five independent runs.

5. Experimental Setup

5.1. Evaluation Questions and Protocol

The experiments address seven questions: whether peptide sequence retains tissue-associated signal after matching the MHC and source protein; the performance of the human model; the contribution of major model components; the effect of excluding every held-out peptide from all training combinations; replication in mouse; the signal recoverable without tissue identity; and whether differences between model structures persist when exact peptide identities are separated. These questions connect the technical comparisons to their biological interpretation. The human–mouse analysis concerns replication of the benchmark definition, not transfer of one model between species.

Many model choices were explored on the ordinary human benchmark. To avoid further adjustment after observing the harder results, the peptide-separated analysis was specified in advance, including the split, model versions, random initializations, training budget, comparison models, performance measures, and statistical tests. Exact split files, configurations, and timestamps are archived with the predictions.

The two main biological questions are not interchangeable. The ordinary internal evaluation asks whether the model can rank held-out pairs within familiar tissue–MHC combinations while allowing an individual peptide to appear in another training combination. The peptide-separated analysis asks whether ranking remains possible when each held-out peptide is absent from every fitting combination. Table 5 defines the corresponding data pools, overlap rules, fold construction, and model repetitions.

For every trainable method, held-out examples are excluded from parameter fitting. In the peptide-separated analysis, a held-out peptide and every pair connected to it remain outside the corresponding fitting fold. A reported prediction therefore cannot benefit from direct exposure to that peptide identity during fitting. This within-run separation does not imply that the overall model design was chosen without earlier access to the ordinary benchmark; model development had already used pair-preserving cross-validation on the same training pool.

To assess sensitivity to peptide reuse, an ordinary reference split is generated from exactly the same pair pool with nearly the same held-out size per tissue–MHC combination. Its intended difference from the peptide-separated split is whether an exact peptide may cross the fitting boundary. Grouping pairs linked directly or indirectly by shared peptides can also alter parent-protein and motif distributions, however. The observed difference is therefore interpreted as a robustness gap under peptide separation rather than a causal estimate of the effect of repeated peptide identity.

5.2. Training and Baseline Models

All encoders and combination-specific output layers are optimized with AdamW [16] for 25 epochs, using a batch size of 512, a learning rate of 10^{-3}, weight decay of 10^{-4}, and gradient-norm clipping at 1.0. The global and HLA-specific pathways are trained independently for each random initialization. Within every cross-validation analysis, only the fitting partition is used to estimate model parameters; held-out rows are used solely for prediction and evaluation.

The human model uses three random initializations: 20260704, 20260705, and 20260706. Its architecture, optimization settings, auxiliary-loss weights, score-combination rule, and random initializations were specified before the reported evaluation. Training follows a prespecified 25-epoch schedule without early stopping on held-out data. Preprocessing and combination mappings are reconstructed from each fitting partition so that held-out labels do not influence training.

The comparison models range from simple sequence rules to architectures that share information across tissue–MHC combinations. One-hot logistic regression provides a linear sequence baseline. A BLOSUM62 random forest incorporates established similarities among amino-acid substitutions. The shared-encoder model learns one peptide representation with a separate output for each tissue–MHC combination. Dual-branch models compare global sharing with sharing restricted to one HLA allele. All methods within a table use the same tissue–MHC combinations and data split.

The BLOSUM62 random forest [17] concatenates the 20 substitution scores at each of nine positions into 180 input features. Each tissue–MHC combination receives a separate forest of 200 trees, with square-root feature subsampling and a minimum leaf size of two. The dual-branch neural controls use 16-dimensional residue vectors, flatten the nine positions, and apply two 128-unit multilayer-perceptron layers with rectified linear activation and dropout of 0.2 after the first layer. The model with auxiliary supervision adds tissue and HLA losses weighted 0.1 to the global pathway; the model without auxiliary supervision retains the same two pathways but omits those losses. Human neural predictions are averaged across three initializations, and mouse predictions in the peptide-separated analysis are averaged across five.

Every model in the peptide-separated comparison uses the same archived fold assignments rather than regenerating the split. The human comparison contains 73,798 pairs in 17,036 groups linked directly or indirectly by shared peptides across 157 tissue–HLA combinations; the mouse comparison contains 6,766 pairs in 1,844 such groups across 24 tissue–H2 combinations. Both use three folds with zero complete-pair and exact-peptide overlap between fitting and held-out partitions.

5.3. Controls Without Tissue Information

The most direct biological control removes tissue identity while retaining peptide and MHC information. If this control

Table 7: Label-conflict audit for controls without tissue information. A peptide–MHC query is conflicting when its retained tissue-specific rows contain both labels within the stated partition. Percentages in the query and row columns use all unique queries and all rows in that partition as their respective denominators. The trainable tissue-blind control preserves these rows and labels rather than collapsing them to a consensus target.

Species	Partition	Rows	Unique queries	Conflicting queries (%)	Rows in conflicting queries (%)
Human	Training pool	147,596	72,136	21,166 (29.34%)	60,137 (40.74%)
Human	Saved internal test	31,400	24,015	1,375 (5.73%)	3,166 (10.08%)
Human	Complete benchmark	178,996	79,759	26,700 (33.48%)	79,281 (44.29%)
Mouse	Training pool	13,532	5,907	2,151 (36.41%)	5,652 (41.77%)
Mouse	Saved internal test	4,800	3,306	381 (11.52%)	859 (17.90%)
Mouse	Complete benchmark	18,332	6,663	3,516 (52.77%)	9,771 (53.30%)

performed as well as the full model, the benchmark could be explained largely by general peptide–MHC compatibility. The human and mouse controls use nearly the same model capacity and exactly the same evaluation splits as their full counterparts. The control is $s = f(p, m)$: it receives peptide sequence and MHC restriction but no tissue input, tissue embedding, tissue auxiliary loss, or tissue–MHC-specific output head.

Rows are not pooled, deduplicated, or assigned a consensus label after tissue identity is removed. Each original tissue-specific row and label is retained in the loss. Consequently, the same peptide–MHC query can carry both labels across tissues: it is positive where ligand evidence was reported and may be a pseudo-negative elsewhere. Table 7 quantifies this deliberate label ambiguity. A "conflicting query" is a peptide sequence and MHC restriction with both labels in the stated partition; a "conflicting row" is any row belonging to such a query. These are not contradictions within one tissue–MHC task. They arise across tissues because the control is intentionally denied the variable that distinguishes those contexts.

We also score every peptide–MHC query with MHCflurry and NetMHCpan, two established predictors that do not use the target tissue [7, 6]. These tissue-agnostic controls quantify how much of the matched-pair benchmark can be explained by general binding or presentation propensity. They are not retrained on the TissuePMHC labels.

The control set contains 79,759 unique human peptide–MHC queries across 35 HLA alleles and 6,663 mouse queries across four H2 restrictions. The MHCflurry presentation score and the NetMHCpan eluted-ligand and binding-affinity scores cover every row and pair. Controls are evaluated on the saved internal test set and on the complete training pools used for the ordinary and peptide-separated cross-validation analyses. Because these external scores are predetermined rather than fitted within each fold, their standalone metrics are identical across the two cross-validation partitions of the same pool.

The high conflict fractions in the complete benchmarks show why this control is intentionally difficult: identical observed inputs can correspond to different tissue-specific targets. Its performance measures how much ranking signal is recoverable from peptide and MHC alone under that ambiguity; it is not an estimate from a noise-free, tissue-pooled presentation dataset.

Finally, we test whether a tissue-agnostic presentation score adds information beyond the tissue-conditioned score. The combination rule is learned on other folds and applied to the held-out fold by cross-fitted stacking. This is a diagnostic analysis rather than a newly proposed final model.

5.4. Performance Metrics and Statistical Analysis

The main question is whether peptides reported in the target tissue rank above comparison peptides with the same MHC restriction and parent protein. AUROC is calculated separately for every tissue–MHC combination and then averaged with equal weight, preventing data-rich tissues or alleles from dominating:

$$\mathrm{MacroAUROC} = \frac{1}{|\mathcal{T}_{\mathrm{valid}}|} \sum_{\tau \in \mathcal{T}_{\mathrm{valid}}} \mathrm{AUROC}_{\tau},$$

where $\mathcal{T}_{\mathrm{valid}}$ contains the prespecified combinations with both labels available. AUPRC is averaged in the same manner:

$$\mathrm{MacroAUPRC} = \frac{1}{|\mathcal{T}_{\mathrm{valid}}|} \sum_{\tau \in \mathcal{T}_{\mathrm{valid}}} \mathrm{AUPRC}_{\tau}\,.$$

Because the paired dataset has a constructed 1:1 label ratio, AUPRC is interpreted relative to the benchmark prevalence within each combination and is not compared directly with AUPRC from naturally imbalanced clinical cohorts.

An average can hide weak performance in difficult tissues, so we also report the ten lowest-performing human combinations and six lowest-performing mouse combinations. Within-pair ordering accuracy is the fraction of matched pairs for which the recorded-positive peptide receives the higher score. Accuracy, F1, and MCC are secondary because they require a single decision threshold and the final score is not a calibrated biological probability.

For each random initialization, predictions from all held-out folds are first concatenated. Metrics are then calculated for each tissue–MHC combination from its complete cross-validation prediction vector and averaged with equal weight. Prediction averaging in the peptide-separated analysis uses only models trained on the corresponding fitting fold. Fold-specific metrics are descriptive and are not treated as independent experimental replicates. Variation across initializations is reported separately from the final averaged prediction.

Uncertainty intervals are obtained by resampling complete tissue–MHC combinations rather than treating folds or repeated runs as independent biological samples. Paired tests compare corresponding combinations, and multiple comparisons are corrected within each planned analysis family. Because combinations may share a tissue, allele, proteins, or related peptides, we emphasize effect size, the numbers improving or declining, and agreement between species rather than treating them as independent patient cohorts.

5.5. Cross-species Interpretation and Reproducibility

Human and mouse use the same matched-pair definition, eligibility threshold, saved internal test design, pair-preserving cross-validation, and peptide-separated cross-validation. They differ in the number of tissue–MHC combinations, MHC system, and data composition. Concordant findings support replication of benchmark learnability but do not establish a species-independent molecular mechanism or a controlled comparison of architectures.

The human model uses random initializations 20260704–20260706; mouse neural predictions in the peptide-separated comparison additionally use 20260707 and 20260708. Both cross-validation schemes use recorded split random state 20260711. Cross-species comparisons emphasize above-random signal and the value of structured information sharing. Numerical differences are not interpreted as species effects because data scale, biological diversity, and model selection differ.

For reproducibility, each experiment records the train/test access policy, fold-assignment manifest, split random state, training initializations, model configuration, and command-line arguments. Reports include epoch count, batch size, optimizer, learning rate, weight decay, dropout, parameter count, training time, peak memory where available, hardware, and software versions. Predictions are saved at row level with tissue–MHC combination, label, pair identifier, fold, model member, and final score.

Dataset and configuration files are accompanied by cryptographic hashes, and each result directory records the timestamped protocol and code state. Metadata for the mouse saved internal test set state that it was accessed only after model selection. Peptide-separated runs retain their peptide-linked-group assignments and zero-overlap audits.

5.6. Data and Code Availability

The accompanying repository contains processed benchmark tables, pair and peptide-linked-group fold manifests, row-level predictions, configuration files, cryptographic hashes, and analysis scripts. Raw material derived from the IEDB and third-party predictor assets remains subject to original access and licensing terms; NetMHCpan executables are not redistributed.

6. Results

6.1. Research Question 1: Is Tissue-specific Preference Learnable?

We first ask whether peptide sequence contains useful tissue-associated signal after holding the MHC restriction and source protein constant. Every retained human or mouse tissue–MHC combination contains more than 200 eligible matched peptide pairs. On the resulting 157-combination human benchmark, the shared-encoder model reaches a mean AUROC of 0.7972 and AUPRC of 0.7829 (Table 8), well above random ordering. More structured information sharing increases AUROC to 0.8239–0.8448. Thus, peptides matched by MHC restriction and parent protein are not exchangeable at random, although the observed patterns may reflect antigen processing, protein abundance, cell composition, or study structure.

Human performance after excluding each held-out peptide from all training combinations is reported with the matched ordinary same-pool reference and uncertainty under Research Question 4 (Table 11); the identical-fold mouse control comparison is reported under Research Question 7.

6.2. Research Question 2: Main Human Benchmark Performance

On the ordinary 157-combination human benchmark, TissuePMHC reaches mean AUROC/AUPRC values of 0.8448/0.8348. Peptides reported in the target tissue generally receive higher scores than their MHC- and parent-protein-matched comparisons, although performance varies markedly across tissue–HLA combinations. Accuracy is 0.7772, MCC is 0.5545, and the mean AUROC of the ten most difficult combinations is 0.6834. Table 8 compares models on the same combination set.

Relative to the directly comparable multilayer perceptron averaged across three runs, TissuePMHC improves mean AUROC by 0.0103 and the difficult-combination summary by 0.0168. The improvement is visible in the mean, median, and lower-performing combinations. Larger differences from the shared-encoder model are descriptive because the aggregation procedures are not identical.

We also repeat the comparison using cross-validation that keeps both rows of every pair in the same fold. This analysis asks whether the advantage extends beyond one saved test set. The closest comparison uses the same two-pathway design with a simpler peptide representation.

The peptide-separated result is reported under Research Question 4 together with its ordinary same-pool reference, a per-combination scatter plot, uncertainty, and four performance measures.

6.3. Research Question 3: Component Contributions on the Human Benchmark

Table 10 examines whether position-aware filters improve performance, whether global and HLA-specific sharing provide complementary rankings, and whether averaging independent runs improves stability. These are completed system comparisons rather than a full factorial ablation. Replacing the simpler peptide representation with a multi-kernel encoder raises mean AUROC from 0.8345 to 0.8448 and the difficult-combination summary from 0.6666 to 0.6834.

The global pathway performs better on average than the HLA-specific pathway, but their combined within-combination ranking reaches 0.8225–0.8249 AUROC per run, above either pathway alone. Allele-restricted learning therefore contributes useful ordering information for some combinations.

Across three independent runs, TissuePMHC obtains mean AUROCs of 0.8345, 0.8314, and 0.8330, with mean 0.8329 and sample standard deviation 0.0016. Averaging their predictions raises AUROC to 0.8448.

Table 8: Human saved internal test benchmark across 157 tissue–HLA combinations. Mean rows summarize three separately trained models; averaged-model rows are calculated after averaging row-level predictions. Only the two averaged-model rows use directly comparable aggregation.

Model	Mean AUROC	Median AUROC	AUPRC	Acc.	MCC	Worst-10
Shared encoder, mean of three runs	0.7972	0.8068	0.7829	0.7277	0.4598	0.6299
Dual branch with auxiliary supervision, mean of three runs	0.8239	0.8351	0.8117	0.7552	0.5132	0.6501
Two-view multilayer perceptron, prediction average	0.8345	0.8441	0.8233	0.7655	0.5313	0.6666
TissuePMHC, prediction average	**0.8448**	**0.8585**	**0.8348**	**0.7772**	**0.5545**	**0.6834**

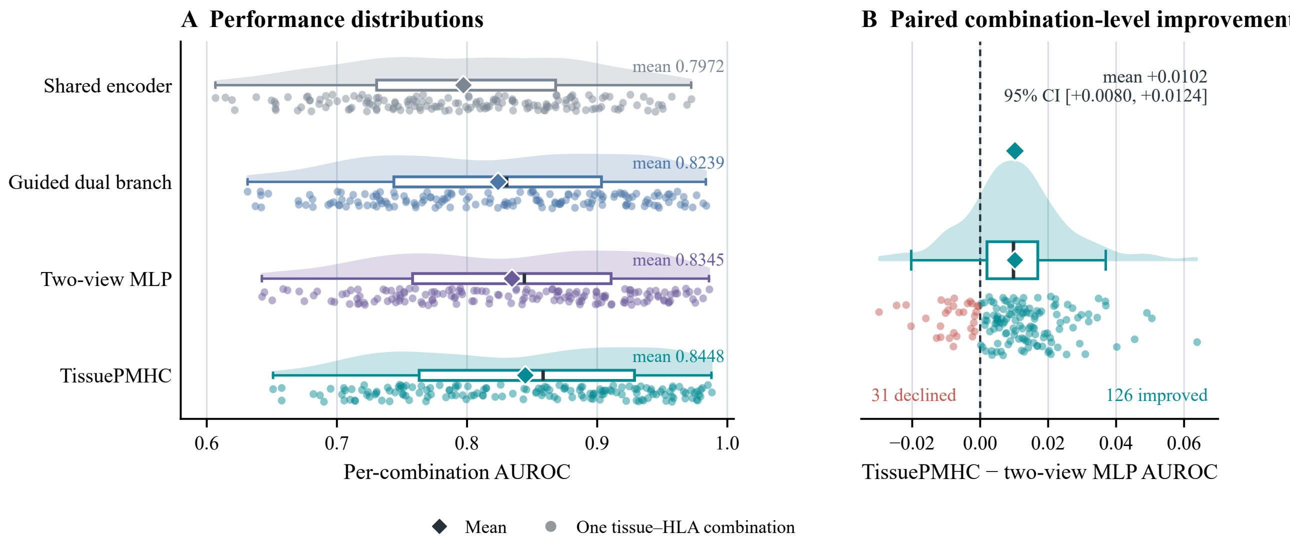

Figure 5: Human performance on the ordinary benchmark. **A**, Raincloud plots show the complete distribution across 157 tissue–HLA combinations: shaded densities summarize distribution shape, points show individual combinations, boxes show the interquartile range and median, and diamonds show the mean. The first two distributions average combination-level AUROCs across three separately trained models; the last two use averaged row-level predictions, matching Table 8. **B**, Paired combination-level AUROC differences between TissuePMHC and the directly comparable two-view multilayer perceptron. The diamond and horizontal interval show the mean difference and its 95% combination-resampling bootstrap interval (10,000 resamples).

Table 9: Human comparison on identical pair-preserving cross-validation folds. This analysis is distinct from the ordinary same-pool reference used to quantify sensitivity to peptide separation.

Model	AUROC	AUPRC	Worst-10
dual-branch model with auxiliary supervision on identical folds	0.8131	0.7980	0.6419
TissuePMHC	**0.8279**	**0.8144**	**0.6562**

Together, the ordinary-benchmark runs provide descriptive support for the assembled design but not a causal decomposition of its components. The identical-fold comparisons under Research Question 7 are stronger: auxiliary tissue and allele supervision retains a clear benefit after peptide separation, whereas neither rank fusion without auxiliary supervision nor the multi-kernel encoder shows a stable advantage over the strongest simpler guided model.

6.4. Research Question 4: Peptides Absent from All Training Combinations

The ordinary reference repartitions the complete training pair pool. Within each tissue–MHC combination, complete pair identifiers are shuffled using the recorded split random state and assigned round-robin to three folds. This keeps both rows of a pair together but allows a held-out peptide to reappear in fitting rows from another tissue–MHC combination. The peptide-separated analysis starts from the same pair pool but assigns entire groups linked through shared peptide sequences to one fold, ensuring that every held-out peptide is absent from all fitting combinations. Both analyses retain familiar tissue–MHC combinations, evaluate the same pooled rows, and have nearly identical held-out sizes. Their difference measures sensitivity to exact-peptide separation within known biological contexts; it does not test unseen tissues, alleles, combinations, or source proteins.

6.4.1. Human benchmark

Removing peptide reuse makes the human ranking problem clearly harder (Table 11). Both analyses evaluate the same 73,798 pairs. Among 471 combination-by-fold cells, 333 have identical held-out sizes and the remainder differ by one pair, giving a mean absolute difference of 0.293. In the ordinary folds, 69.19–69.91% of unique held-out peptides reappear during fitting, compared with 0% after peptide separation.

Figure 6 compares the two AUROC values for each human tissue–HLA combination.

Table 10: Evidence about system components from completed runs on the 157-combination human benchmark. The comparisons do not constitute a factorial ablation, and no row isolates a component effect under peptide-separated evaluation.

Design question	Completed comparison	AUROC	Descriptive Δ
Multi-kernel peptide encoding	Multilayer perceptron → TissuePMHC; predictions averaged across three runs	0.8345 → 0.8448	+0.0103
Pathway complementarity	Better single pathway → combined within-combination ranking; ranges across runs	0.8072–0.8091 → 0.8225–0.8249	+0.0134–0.0177
Prediction averaging	Mean of separate runs → average of three prediction sets	0.8329 → 0.8448	+0.0118

Table 11: Human AUROC under same-pool ordinary cross-validation and cross-validation excluding each held-out peptide from all fitting combinations. Difference is peptide-separated minus ordinary same-pool cross-validation; a negative value denotes lower performance after exact peptide separation.

Ordinary cross-validation	Peptide-separated cross-validation	Difference
0.8280	0.7652	-0.0628

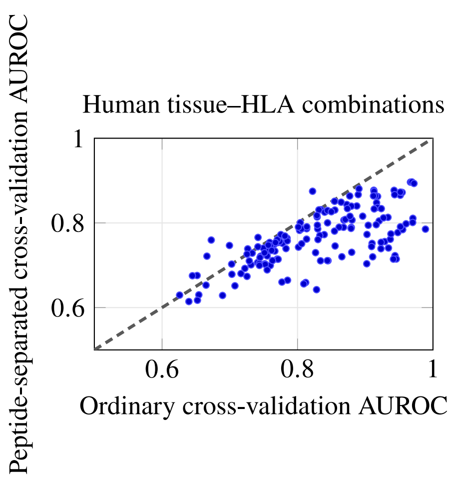

Figure 6: Human AUROC for each tissue–HLA combination under ordinary same-pool and peptide-separated cross-validation. Points below the gray equality line have lower performance after exact-peptide separation.

Mean human AUROC decreases by 0.0628 and is lower for 145 of 157 tissue–HLA combinations. The 95% interval obtained by resampling complete combinations is -0.0728 to -0.0534. Within-pair ordering accuracy decreases by 0.0529. Exact peptide reuse therefore explains part, but not all, of the easier performance estimate.

6.4.2. Mouse benchmark

The mouse peptide-separated benchmark uses the same 6,766 pairs, grouped into 1,844 peptide-linked components with zero held-out-peptide overlap. A matched ordinary-versus-separated estimate is not reported here because the retained mouse controls were not all evaluated under a common ordinary-fold protocol. Their identical-fold peptide-separated comparison is reported under Research Question 7.

Saved-test results are not subtracted from peptide-separated cross-validation results because the evaluations contain different examples.

The peptide-separated analysis removes exact-peptide reuse while retaining familiar tissue–MHC combinations and other shared context. Table 5 and the Limitations define its scope.

6.5. Research Question 5: Replication in Mouse Tissue–MHC (H-2) Combinations

The mouse benchmark asks whether the same MHC- and parent-protein-controlled comparison is learnable across 24 tissue–H2 combinations. Mouse models are trained independently, and no human parameters are transferred.

Table 12 adds the completed mouse implementations of method families evaluated in the human benchmark. Every row uses the same fixed mouse training and test partitions, and every metric gives equal weight to the 24 combinations. The H2 pseudo-sequence-conditioned and ID–pseudo-sequence hybrid models are omitted because no provenance-verified H2 pseudo-sequence file was available. The full multi-kernel TissuePMHC aggregate is numerically strongest, followed by the MLP dual-branch aggregate. Because multiple architectures have been compared retrospectively on the saved test set, this survey is an internal benchmark comparison rather than untouched confirmation or external validation.

On the saved internal test set, the strongest retained prediction aggregate has an equal-weight AUROC/AUPRC of 0.8528/0.8477. Because the same saved benchmark was used for the retrospective method survey, this value characterizes the observed benchmark rather than providing independent validation. Peptide-separated results for the matched mouse controls are reported under Research Question 7.

6.6. Research Question 6: How Much Signal Remains without Tissue Identity?

Removing tissue identity while retaining peptide and MHC information lowers human performance in every evaluation. Under peptide-separated cross-validation, AUROC falls from 0.7652 to 0.6630, with decreases in 156 of 157 combinations. General peptide–MHC compatibility therefore does not reproduce the complete ranking signal, although this does not prove a causal tissue-processing mechanism.

This comparison must be interpreted together with the conflict audit in Table 7. Once tissue is hidden, an identical peptide–MHC input can retain different labels from different tissue contexts. The performance decrease therefore reflects the information lost with tissue identity, including the resulting input–label ambiguity; it should not be attributed solely to a molecular tissue-processing effect.

MHCflurry and NetMHCpan perform better than random on the human matched pairs (Table 14). General binding and presentation propensity therefore contribute useful informa-

Table 12: Expanded retrospective method survey on the mouse saved internal test set across 24 tissue–H2 combinations. Ordinary rows are arithmetic means of three runs (seeds 20260704–20260706); rows marked † aggregate row-level predictions across the stated runs. Worst-6 is the mean AUROC of the six lowest-performing combinations within each run or prediction aggregate.

Model	AUROC	AUPRC	Worst-6	Evidence
Neural single-task MLP	0.7315	0.7078	0.6424	three-run mean
Tissue-grouped hard sharing	0.7329	0.7104	0.6460	three-run mean
Conditioned tissue and H2 IDs	0.8158	0.8118	0.6689	three-run mean
Selective global/H2 grouping	0.8258	0.8192	0.6907	three-run mean
DB-MTL shared heads	0.8301	0.8207	0.6942	three-run mean
MMoE, 4 experts × width 256	0.8309	0.8227	0.6963	three-run mean
MMoE, 6 experts × width 128	0.8315	0.8268	0.7023	three-run mean
Validation-softmax dual-branch ensemble	0.8329	0.8275	0.6965	three-run mean
Shared encoder with task heads	0.8333	0.8254	0.6965	three-run mean
Validation-delta clipped dual-branch ensemble	0.8355	0.8295	0.7005	three-run mean
Fixed global/H2 dual-branch average	0.8372	0.8304	0.7031	three-run mean
Auxiliary-global + auxiliary-H2 dual branch	0.8400	0.8340	0.7057	three-run mean
Auxiliary-global + plain-H2 dual branch	0.8415	0.8356	0.7068	three-run mean
MLP dual-branch rank average†	0.8488	0.8436	0.7153	three-run aggregate
Full multi-kernel TissuePMHC†	**0.8528**	**0.8477**	**0.7196**	three-run aggregate

Table 13: Mouse saved-test AUROC grouped by H2 restriction. Values are descriptive subgroup means rather than controlled restriction effects.

H2 restriction	H2-Db	H2-Kb	H2-Kd	H2-Kk
Mean AUROC across tissues	0.9152	0.7404	0.8397	0.8024

tion. The full tissue-conditioned model nevertheless has the highest value for every reported metric and evaluation.

The NetMHCpan eluted-ligand score is consistently at least as strong as its binding-affinity score. The AUROC difference between eluted-ligand score and binding-affinity score is 0.0256 on the human saved internal test set and 0.0264 on the human complete training pool. We therefore use the eluted-ligand score as the primary NetMHCpan control and report the complete binding-affinity score sensitivity analysis in Table A4.

The human tissue-conditioned model remains higher than the strongest tissue-agnostic control in every evaluation. AUROC is 0.8448 versus 0.6851 on the saved test, 0.8279 versus 0.6871 under ordinary same-pool cross-validation, and 0.7652 versus 0.6871 after peptide separation; the peptide-separated advantage is 0.0781.

Cross-fitted stacking yields only marginal improvement over the human tissue-conditioned score alone. The largest change is observed for peptide-separated cross-validation with MHCflurry, from 0.7652 to 0.7693 AUROC; the saved-test value changes from 0.8448 to 0.8453. Complete results appear in Table A5.

6.7. Research Question 7: Model Structure under Peptide Separation

Every model is retrained on the same peptide-separated folds. All human models retain useful signal (Table 15). TissuePMHC clearly exceeds simple sequence baselines and modestly exceeds the shared encoder and unguided dual-branch models.

The saved-test continuation broadens the retrospective survey without changing the protocol-specific conclusions. The full TissuePMHC prediction aggregate is numerically strongest, followed by the MLP dual-branch aggregate. Among ordinary three-run means, the auxiliary-global plus plain-HLA dual branch is strongest. These comparisons reuse the saved benchmark and therefore support method characterization rather than independent model selection.

Across the 15 shared algorithm families in Figure 7, the mouse saved-test AUROC is numerically higher by 0.0080–0.0481. The ordering is nevertheless broadly consistent: dual-branch prediction aggregates occupy the strongest end in both species, whereas single-task and tissue-grouped training are weaker. Dataset size, task composition, allele inventory, and test examples differ, so the numerical gaps cannot be attributed to species.

Comparison with the strongest simpler human control warrants caution. The guided dual-branch model uses the same two-pathway structure with a simpler peptide representation. TissuePMHC is only 0.0010 AUROC higher, with 76 positive, one tied, and 80 negative per-combination differences; the uncertainty interval includes zero. By contrast, adding tissue and HLA supervision to the unguided dual branch improves AUROC by 0.0087 and benefits 144 of 157 combinations. The clearest retained effect is therefore biological-group supervision rather than the more elaborate encoder alone.

The mouse comparison matches the human strict table for one-hot logistic regression, BLOSUM62 random forest, shared encoding, and the two dual-branch controls (Table 16). The auxiliary dual branch has the highest mean AUROC (0.7590). Auxiliary supervision improves the matched dual branch by 0.0034 AUROC and is higher in 21 of 24 combinations (95% interval, 0.0023–0.0047). Mean AUROC differs by at most 0.0083 among the random forest,

Table 14: Human comparison of controls that omit tissue identity with the full tissue-conditioned TissuePMHC model. The control retaining peptide and MHC information has approximately the same capacity as the full model. MHCflurry uses its predicted-presentation score and NetMHCpan its eluted-ligand score. External scores are predetermined, so their values are identical across the two cross-validation partitions of the same pool.

Evaluation metric	Peptide + MHC	MHCflurry presentation	NetMHCpan eluted-ligand score	**Full tissue-conditioned**
Human				
Saved internal test: AUROC	0.7449	0.6851	0.6833	**0.8448**
Saved internal test: AUPRC	0.7563	0.6783	0.6686	**0.8348**
Saved internal test: within-pair accuracy	0.7243	0.6886	0.6897	**0.8375**
Same-pool ordinary cross-validation: AUROC	0.7353	0.6871	0.6823	**0.8279**
Same-pool ordinary cross-validation: AUPRC	0.7379	0.6751	0.6632	**0.8144**
Same-pool ordinary cross-validation: within-pair accuracy	0.7144	0.6898	0.6875	**0.8229**
Peptide-separated cross-validation: AUROC	0.6630	0.6871	0.6823	**0.7652**
Peptide-separated cross-validation: AUPRC	0.6502	0.6751	0.6632	**0.7452**
Peptide-separated cross-validation: within-pair accuracy	0.6691	0.6898	0.6875	**0.7700**

Table 15: Human architecture comparison across 157 tissue–HLA combinations on identical peptide-separated folds. Neural methods average row-level predictions across runs; logistic regression has no random training variation.

Model	Mean AUROC	Median	AUPRC	Within-pair accuracy	Worst-10	Worst combination
One-hot logistic regression	0.7127	0.7069	0.6906	0.7157	0.5873	0.5533
BLOSUM62 random forest	0.7212	0.7208	0.7007	0.7263	0.5895	0.5634
Shared encoder	0.7561	0.7565	0.7340	0.7631	0.6248	0.6003
Dual branch without auxiliary supervision	0.7555	0.7577	0.7339	0.7575	0.6163	0.5905
Dual branch with auxiliary supervision	0.7642	**0.7660**	0.7433	0.7686	0.6283	0.6034
TissuePMHC	**0.7652**	0.7606	**0.7452**	**0.7700**	**0.6382**	**0.6142**

shared encoder, and two dual-branch models. Thus, peptide-separated mouse performance supports learnability and a small benefit from auxiliary supervision rather than broad architectural superiority.

Prediction averaging materially improves the final neural estimates. Relative to the mean of separate runs, AUROC gains are +0.0248 for the human shared encoder, +0.0161 for the unguided dual branch, +0.0139 for the guided dual branch, and +0.0147 for TissuePMHC. Mouse gains are +0.0211 for the shared encoder, +0.0175 for the unguided dual branch, and +0.0167 for the auxiliary dual branch, compared with +0.0016 for the random forest and zero for one-hot logistic regression.

Table 15: Human architecture comparison (continued): retrospective survey on the saved internal test set. These rows use the 157 tissue–HLA combinations in the saved-test benchmark and are not directly comparable to the peptide-separated rows in the preceding panel. Ordinary neural rows are arithmetic means of three runs; rows marked † aggregate row-level predictions.

Model	AUROC	AUPRC	Acc.	MCC	Worst-10	Evidence
One-hot logistic regression	0.7238	0.7065	0.6636	0.3288	0.5939	single fit
BLOSUM62 logistic regression	0.7227	0.6985	0.6672	0.3357	0.5950	single fit
BLOSUM62 HistGradientBoosting	0.7240	0.7066	0.6668	0.3351	0.5893	single fit
BLOSUM62 Extra Trees	0.7279	0.7120	0.6680	0.3386	0.5958	single fit
BLOSUM62 random forest	0.7297	0.7146	0.6707	0.3444	0.5969	single fit
Neural single-task MLP	0.6934	0.6803	0.6410	0.2841	0.5682	three-run mean
Shared encoder with task heads	0.7972	0.7829	0.7277	0.4598	0.6299	three-run mean
Conditioned tissue and HLA IDs	0.7955	0.7856	0.7309	0.4648	0.5662	three-run mean
HLA pseudo-sequence conditioned	0.7866	0.7761	0.7258	0.4549	0.5371	three-run mean
HLA ID + pseudo-sequence hybrid	0.7963	0.7867	0.7334	0.4701	0.5667	three-run mean
FAMO shared heads	0.7668	0.7518	0.7038	0.4103	0.5997	three-run mean
HLA-grouped hard sharing	0.7972	0.7855	0.7347	0.4716	0.5942	three-run mean
Tissue-grouped hard sharing	0.7122	0.6946	0.6549	0.3122	0.5878	three-run mean
Selective global/HLA grouping	0.8035	0.7902	0.7377	0.4788	0.6137	three-run mean
Fixed global/HLA dual-branch average	0.8184	0.8069	0.7513	0.5055	0.6440	three-run mean
Validation-delta clipped ensemble	0.8175	0.8061	0.7495	0.5019	0.6354	three-run mean
Validation-softmax ensemble	0.8136	0.8020	0.7458	0.4947	0.6259	three-run mean
MMoE	0.8070	0.7930	0.7381	0.4822	0.6379	three-run mean
MMoE, 4 experts × width 256	0.8076	0.7945	0.7388	0.4833	0.6271	three-run mean
MMoE, 6 experts × width 128	0.8050	0.7920	0.7356	0.4759	0.6319	three-run mean
DB-MTL shared heads	0.7820	0.7672	0.7172	0.4369	0.5972	three-run mean
Pair-ranking objective	0.7849	0.7692	0.7209	0.4438	0.6059	three-run mean
Tissue/HLA auxiliary supervision	0.8082	0.7921	0.7353	0.4772	0.6474	three-run mean
Auxiliary-global + plain-HLA dual branch	0.8239	0.8117	0.7552	0.5132	0.6501	three-run mean
Auxiliary-global + auxiliary-HLA dual branch	0.8202	0.8082	0.7510	0.5056	0.6502	three-run mean
MLP dual-branch rank average†	0.8345	0.8233	0.7655	0.5313	0.6666	three-run aggregate
Full multi-kernel TissuePMHC†	**0.8448**	**0.8348**	**0.7772**	**0.5545**	**0.6834**	three-run aggregate

Table 16: Mouse architecture comparison across 24 tissue–H2 combinations on identical peptide-separated folds. All entries average predictions across five runs.

Model	Mean AUROC	Median AUPRC		Within-pair accuracy	Worst-6	Worst combination
One-hot logistic regression	0.7335	0.7395	0.7050	0.7363	0.6266	0.5709
BLOSUM62 random forest	0.7507	0.7768	0.7295	0.7573	0.6418	**0.5819**
Shared encoder	0.7538	0.7786	0.7321	0.7567	0.6430	0.5477
Dual branch without auxiliary supervision	0.7556	0.7758	0.7372	0.7542	0.6416	0.5496
Dual branch with auxiliary supervision	**0.7590**	**0.7797**	**0.7395**	**0.7605**	**0.6464**	0.5494

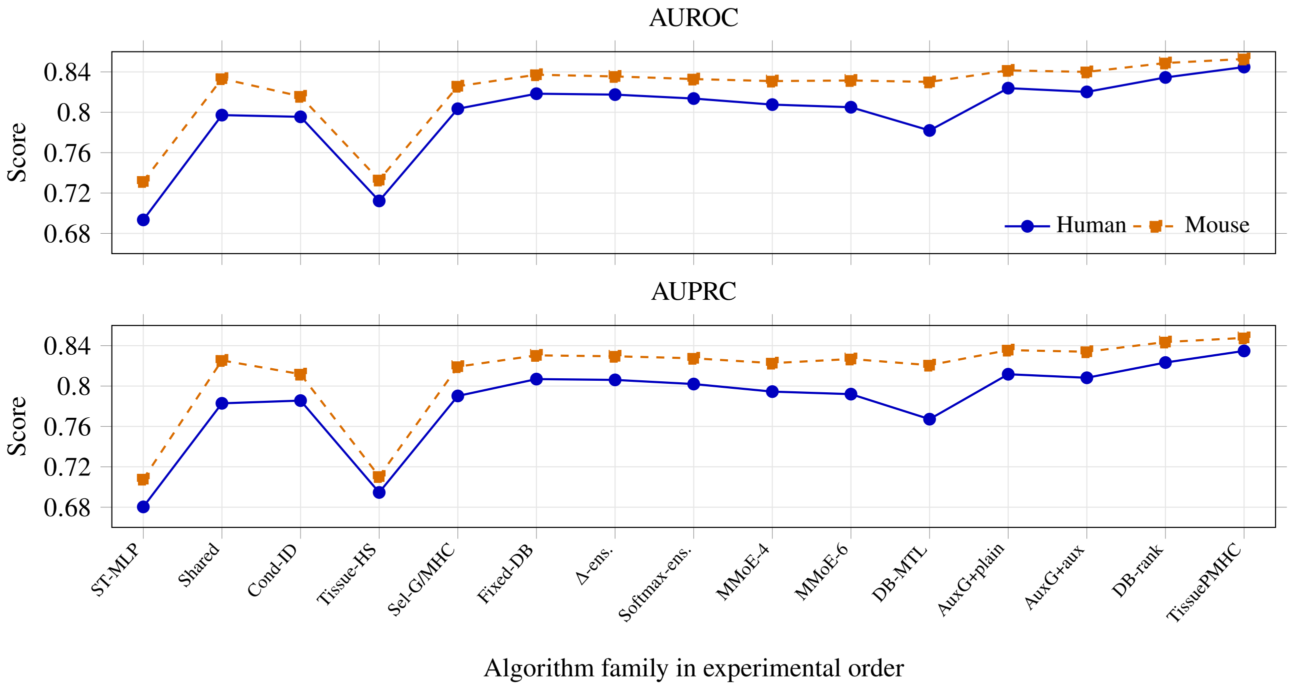

Figure 7: Score trends for matched algorithm families evaluated separately on the human and mouse saved internal test benchmarks. The upper and lower panels show AUROC and AUPRC, respectively. Algorithm families follow the experimental order used in the original comparison; connecting lines guide cross-species trend comparison and do not imply a continuous optimization trajectory. Ordinary points are three-run means, whereas DB-rank and TissuePMHC aggregate row-level predictions across three runs. ST-MLP denotes the neural single-task MLP; Cond-ID, conditioned tissue and MHC IDs; Tissue-HS, tissue-grouped hard sharing; Sel-G/MHC, selective global/MHC grouping; DB, dual branch; ens., ensemble; and AuxG, auxiliary-global. Each species uses its own training data, task inventory, and MHC representation, so the gaps are not estimates of a species effect.

7. Analysis

7.1. Variation among Tissue–HLA Combinations

This section examines variation in prediction difficulty and agreement between the model's two pathways. The observed patterns can motivate biological or data-quality hypotheses but cannot identify a molecular mechanism.

Performance varies substantially across the 157 human tissue–HLA combinations. On the ordinary benchmark, the ten lowest-performing combinations have a mean AUROC of 0.6834, compared with an overall mean of 0.8448. These low values are concentrated in lung, blood, and lymphoid labels and range from 0.6510 for lung HLA-B*49:01 to 0.7056 for lymphoid tissue HLA-B*13:02. Biological similarity, cellular heterogeneity, annotation, sampling, and study effects are all plausible explanations.

Table 17 reports the tissue-level means for all 25 tissues, calculated with equal weight across the available HLA combinations within each tissue. Estimates for tissues represented by only one HLA combination are descriptive and potentially unstable.

Prediction difficulty is not explained simply by the number of training examples. The association between training size and AUROC is weak under both evaluations and is not statistically reliable. A large tissue–HLA combination can remain difficult when tissue composition is heterogeneous, peptide evidence overlaps other tissues, or records originate from diverse studies.

Table 18 provides a descriptive summary by HLA locus for the saved test and peptide-separated analyses. HLA-A has the highest saved-test mean and HLA-C the lowest; HLA-C also shows the smallest descriptive decrease.

These locus summaries combine different alleles, tissues, training sizes, and peptide-linked group structures and therefore do not estimate controlled HLA-locus effects. Differences are calculated from unrounded combination means, so subtracting displayed values can differ by 0.0001. The direct same-pool comparison used to quantify sensitivity to peptide separation appears under Research Question 4 and in Figure A1.

Saved-test and peptide-separated AUROCs correlate moderately across tissue–HLA combinations, with Spearman $\rho = 0.569$ and $p < 10^{-14}$. The descriptive median difference between these evaluations is -0.0652, with 16 increases and 141 decreases. Six of the ten largest decreases involve HLA-A*11:01 and two involve HLA-B*49:01, motivating follow-up that accounts for allele and peptide-linked-group structure without implying intrinsic allele effects.

7.2. Complementarity of the Two Human Model Pathways

We compare the global pathway with the HLA-specific pathway. Their peptide scores are moderately correlated on average, but agreement varies widely across tissue–HLA combinations, indicating related but non-identical sequence information.

The global pathway has the higher overall mean but outperforms the HLA-specific pathway in only 81 of 157 combinations; the HLA-specific pathway performs better in the remaining 76. Global sharing is therefore stronger on average, while allele-restricted learning retains useful combination-specific ordering information.

The combined ranking outperforms the better single pathway in 260 of 471 run-by-combination comparisons. Its median improvement is 0.0017 and mean improvement 0.0003. A simple measure of pathway disagreement, $1 - \mathrm{corr}(s_g, s_h)$, is not significantly associated with the benefit of combination, with Spearman $\rho = -0.062$ and $p = 0.181$. Disagreement may represent either complementary signal or pathway-specific error.

These results support the prespecified equal-weight combination as a robust aggregate rule but do not imply that every tissue–HLA combination benefits from both pathways.

7.3. Error Analysis

Per-combination results point to several recurring sources of difficulty. Low-performing combinations are concentrated in blood and lymphoid categories, where related cell populations and broad sampling may produce overlapping peptide evidence. Combinations with few examples can yield unstable estimates even though training size alone does not explain performance. Peptides reported across several tissues can also become difficult pseudo-negatives when an unob-

Table 17: Descriptive human tissue-level performance on the 157-combination ordinary benchmark. Means are taken with equal weight across the available HLA combinations within each tissue. All 25 tissues are shown; values continue from the left block to the right block in ascending AUROC order.

Tissue	AUROC	AUPRC	Tissue	AUROC	AUPRC
Blood	0.7618	0.7587	Colon	0.9204	0.9135
Lymphoid	0.7914	0.7744	Liver	0.9281	0.9164
Uterine cervix	0.8021	0.7797	Kidney	0.9287	0.9257
Brain	0.8085	0.8012	Thyroid	0.9316	0.9252
Lung	0.8135	0.8081	Bone marrow	0.9345	0.9256
Thymus	0.8562	0.8405	Esophagus	0.9402	0.9327
Bone	0.8585	0.8662	Small intestine	0.9534	0.9515
Lymph node	0.8680	0.8415	Testis	0.9573	0.9597
Central nervous system (CNS)	0.8937	0.8883	Tongue	0.9574	0.9535
Spleen	0.8947	0.8869	Uterus	0.9617	0.9598
Ovary	0.9014	0.8927	Adrenal gland	0.9778	0.9768
Heart	0.9033	0.8798	Umbilical cord blood	0.9791	0.9805
Breast	0.9095	0.9184			

Table 18: Descriptive human AUROC by HLA locus. Difference is peptide-separated cross-validation minus the saved internal test result. Because the evaluated rows differ, this is not an estimate of the peptide-overlap effect.

HLA locus	Comb.	Saved test	Peptide-separated	Diff.
HLA-A	55	0.8688	0.7824	-0.0864
HLA-B	79	0.8380	0.7507	-0.0873
HLA-C	23	0.8106	0.7738	-0.0369

served target-tissue record reflects incomplete measurement rather than true absence. Study and batch structure may also introduce systematic error.

8. Discussion

8.1. Principal Findings

This study formulates tissue-specific MHC-I presentation as a shared-learning problem over peptide pairs matched by MHC restriction and parent protein. Rather than asking whether a peptide binds to or is presented by an MHC molecule in general, the benchmark asks which member of a matched pair has stronger evidence in a particular tissue–MHC combination. This formulation controls two major sources of trivial discrimination and focuses evaluation on tissue-conditioned relative evidence.

Four findings carry the main biological message. First, peptide sequence contains reproducible tissue-associated information after controlling MHC restriction and source protein. Second, this result appears in both human and mouse, although the species use different models and do not constitute a controlled species comparison. Third, performance falls substantially when a test peptide is absent from every training combination, showing that repeated peptide identities make the ordinary benchmark easier. Fourth, removing tissue identity causes a further decrease, whereas general MHC binding and presentation predictors recover only part of the signal. Tissue context therefore matters to the ranking problem, but the sequence-only models cannot distinguish tissue expression, antigen processing, cell composition, and study structure.

The ordinary and peptide-separated analyses answer complementary questions. The ordinary benchmark evaluates new matched pairs in familiar tissue–MHC combinations while allowing peptide reuse elsewhere in training. The harder analysis evaluates exact peptide identities absent from all fitting combinations within those same biological contexts. Neither evaluates a new tissue, new allele, or independent cohort.

8.2. Effects of Model Structure under Peptide Separation

Architectural differences are smaller after peptide separation than on the ordinary benchmark. In human, TissuePMHC remains above simple sequence baselines, the shared encoder, and the unguided dual branch, but is statistically comparable to the simpler guided dual branch. Auxiliary tissue and allele supervision is therefore the most reproducible component benefit; the comparison does not establish an independent advantage of the multi-kernel peptide encoder.

The mouse interpretation is more conservative. In the identical-fold comparison of simple and dual-branch controls, the auxiliary dual branch has the highest mean AUROC. Auxiliary supervision gives a small but consistent improvement over its matched unguided control, while the random forest, shared encoder, and two dual-branch models remain close in absolute performance. The result therefore supports learnability of the biological comparison and a limited benefit from auxiliary supervision rather than a universal architectural advantage.

8.3. Relationship to General Peptide–MHC Prediction

TissuePMHC is not a replacement for general peptide–MHC prediction. General tools estimate whether a peptide is compatible with an MHC molecule or likely to be presented overall. TissuePMHC estimates tissue-associated preference relative to a matched comparison peptide. MHCflurry and NetMHCpan achieve AUROCs of approximately 0.65–0.69, confirming that general presentation propensity is relevant but does not fully explain the tissue-conditioned ranking.

This distinction limits causal interpretation. Tissue-associated sequence patterns may reflect source-protein expression, antigen processing, cell composition, experimental sampling, or residual study structure. Because the model receives only peptide sequence and a tissue–MHC identifier, it cannot identify which process produced an association. Adding a general presentation score changes AUROC by at most 0.0041.

8.4. Potential Applications

The benchmark may support several research applications. Tissue-conditioned scores could help prioritize candidate antigens whose presentation evidence is concentrated in a target tissue, rank potential normal-tissue off-targets during early therapeutic screening, and select peptides for follow-up immunopeptidomics experiments. The same framework could also contribute to candidate triage for T-cell receptor therapies, vaccines, and other immunotherapies when used alongside binding affinity, immunogenicity, expression, safety, and population-coverage evidence.

These are potential uses rather than validated clinical applications. The balanced benchmark does not reproduce clinical prevalence, and retrospective database associations cannot establish safety or therapeutic benefit. Deployment would require prospective validation, calibrated decision thresholds, uncertainty estimates, and representative cohorts. A low predicted score must not be treated as evidence that a peptide is absent from normal tissue.

8.5. Limitations

The benchmark has several important limitations. The negative member of a pair is a pseudo-negative based on missing database evidence, not an experimentally confirmed non-presented peptide. Incomplete tissue sampling and limited detection depth can therefore convert unobserved positives into apparent negatives. This issue is especially relevant for peptides reported across multiple tissues and for tissues with heterogeneous cellular composition.

The source data are observational and inherit biases from studies in the IEDB. Tissue coverage, HLA frequency, sample preparation, mass-spectrometry platform, reporting practice, and detection depth are uneven. Matching on MHC and parent protein reduces specific confounders but does not remove study, laboratory, donor, or batch structure. The deliberately balanced 1:1 benchmark supports controlled ranking comparisons but does not represent real-world prevalence; its AUPRC and score thresholds should not be transferred directly to deployment settings.

Source-tissue text is retained exactly as recorded in the IEDB. Synonyms, anatomical subregions, and differently formatted labels are not normalized to a controlled ontology, so semantically related tissues may remain separate. Resolving those labels would change the benchmark and require retraining all models.

The present work covers unmodified canonical 9-mer MHC-I ligands. It does not address variable-length ligands, post-translationally modified or spliced peptides, or MHC-II presentation. It also excludes tissue transcriptomics, proteomics, source-protein abundance, flanking sequence, cell-type composition, and antigen-processing features. Tissue information enters through the combination definition and parameter-sharing structure rather than direct biological measurements.

Generalization is restricted in several ways. In the ordinary evaluation, peptides and parent proteins can recur across fitting and evaluation data through different tissue–MHC combinations. Peptide-separated cross-validation prevents exact-peptide reuse but still permits recurring parent proteins, studies, platforms, and similar motifs. It does not enforce new parent proteins, new studies, future data, or external cohorts. Moreover, combination-specific output layers restrict all evaluations to previously represented tissue–MHC combinations; unseen tissues and alleles are not tested.

The human saved test set was used repeatedly during a long model-development process, so project-level selection may have adapted to that benchmark in ways not captured by within-run confidence intervals. The mouse saved test was examined only after model selection, but 81.47% of its unique peptides and 88.88% of its UniProt parent identifiers appear in training. It therefore neither prevents all entity overlap nor constitutes external validation.

MHCflurry and NetMHCpan provide tissue-agnostic external scores but are not guaranteed to be free of overlap with their own training data. Their pretraining may include IEDB observations or related immunopeptidomic studies. Peptide separation in the present benchmark therefore cannot guarantee that held-out peptides were absent from external pretraining collections. The near-tie between MHCflurry and the NetMHCpan eluted-ligand score in human is descriptive and does not establish superiority of either predictor.

Finally, benchmark scale, MHC system, data composition, and the evaluated model sets differ between species, so the cross-species results cannot identify species effects. Within each species, identical-fold comparisons show that architectural superiority is not universal: the strongest guided dual-branch human model is comparable to TissuePMHC, and the mouse auxiliary dual branch has only a small advantage over its unguided counterpart.

No control based on randomly reassigned tissue labels or pathway assignments is reported. Such a control requires refitting models after breaking the tissue association and cannot be reconstructed from archived predictions. The models retaining peptide and MHC information but omitting tissue identity answer the narrower question of performance without tissue information.

8.6. Future Work

Several extensions follow directly from these limitations. Data splits that exclude repeated parent proteins or studies, separate future from past observations, or use external cohorts should test whether performance survives removal of additional overlap and provenance shortcuts. Models refitted after randomly reassigning tissue labels or pathways could determine whether tissue-associated gains exceed those available from the combination structure alone. Auditable checks against external predictor training collections would strengthen interpretation of the tissue-agnostic controls.

Second, the input representation can be expanded to variable-length peptides and MHC-II ligands. Tissue transcriptomics, proteomics, source-protein abundance, flanking sequences, proteasomal cleavage features, and cell-type composition could help separate peptide-intrinsic patterns from tissue context. These additions should be evaluated incrementally under splits that prevent inappropriate overlap.

Third, positive–unlabeled learning and explicit models of observation probability may better reflect the fact that database absence is not biological absence. Calibrated uncertainty estimates could identify tissue–MHC combinations and peptides with insufficient evidence. Representations derived from tissue and MHC features, rather than a separate learned output for every observed combination, could enable principled evaluation on unseen tissues or alleles.

Finally, controlled cross-species pretraining and adaptation could test whether learned features transfer beyond the shared benchmark definition. Such experiments require harmonized data and directly comparable baselines.

9. Conclusion

We introduce human and mouse benchmarks that compare peptides from the same parent protein under the same MHC restriction and predict which has stronger presentation evidence in a specified tissue. Peptide sequence supports this ranking in both species, and removing tissue identity substantially reduces human performance. Human performance also decreases when every held-out peptide is excluded from all fitting combinations, showing that repeated peptide identities make estimates in familiar contexts more optimistic. Identical-fold mouse controls retain above-random performance under the same peptide-separation rule. The retained signal supports tissue-associated peptide ranking in previously represented tissue–MHC combinations but does not identify a specific antigen-processing mechanism. Complex architectures do not consistently outperform strong simpler controls after peptide separation. The main contribution is therefore a biologically focused and reproducible benchmark that distinguishes tissue-associated signal from general peptide–MHC compatibility while making the limits of unseen-peptide prediction explicit. Validation in unseen tissues, alleles, studies, and prospective samples remains necessary before biological or clinical application.

Table A1: Paired statistics comparing ordinary and peptide-separated cross-validation within each tissue–MHC combination. Performance loss is calculated as ordinary minus peptide-separated performance, so positive values indicate a decrease after peptide separation. The table reports the numbers of losses, ties, and gains and 95% confidence intervals from 10,000 resamples of complete tissue–MHC combinations; q_{BH} adjusts eight Wilcoxon tests. This comparison is a downstream analysis rather than a new data split.

Human

Metric	Mean loss (median; median shift)	Loss/tie/gain	95% CI	Adjusted q
AUROC	0.0628 (0.0484; 0.0552)	145/0/12	[0.0534, 0.0728]	1.29×10^{-22}
AUPRC	0.0691 (0.0509; 0.0621)	147/0/10	[0.0588, 0.0799]	1.29×10^{-22}
Accuracy	0.0625 (0.0455; 0.0563)	138/2/17	[0.0529, 0.0727]	2.12×10^{-22}
MCC	0.1249 (0.0907; 0.1125)	139/0/18	[0.1057, 0.1453]	2.12×10^{-22}

Mouse

Metric	Mean loss (median; median shift)	Loss/tie/gain	95% CI	Adjusted q
AUROC	0.0863 (0.0832; 0.0859)	22/0/2	[0.0632, 0.1106]	6.81×10^{-7}
AUPRC	0.0994 (0.1040; 0.1002)	22/0/2	[0.0706, 0.1293]	1.67×10^{-6}
Accuracy	0.0918 (0.0804; 0.0902)	23/0/1	[0.0652, 0.1207]	4.77×10^{-7}
MCC	0.1838 (0.1632; 0.1802)	23/0/1	[0.1306, 0.2415]	4.77×10^{-7}

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A. Additional Generalization Diagnostics

Figure A1 examines two generalization diagnostics using the same peptide-pair pools as the fourth research question. The upper panel reports, for each HLA gene group or H2 restriction, the mean AUROC loss after every held-out peptide is excluded from fitting in all combinations, calculated as ordinary minus peptide-separated cross-validation. A positive bar therefore indicates lower performance under peptide separation. The lower panel reports the mean percentage of unique held-out parent proteins that remain present in the fitting partition. Nonzero values show that separating exact peptide sequences does not also guarantee separation by parent protein. The ordinary cross-validation reference uses the same pooled rows and is distinct from the saved internal test set.

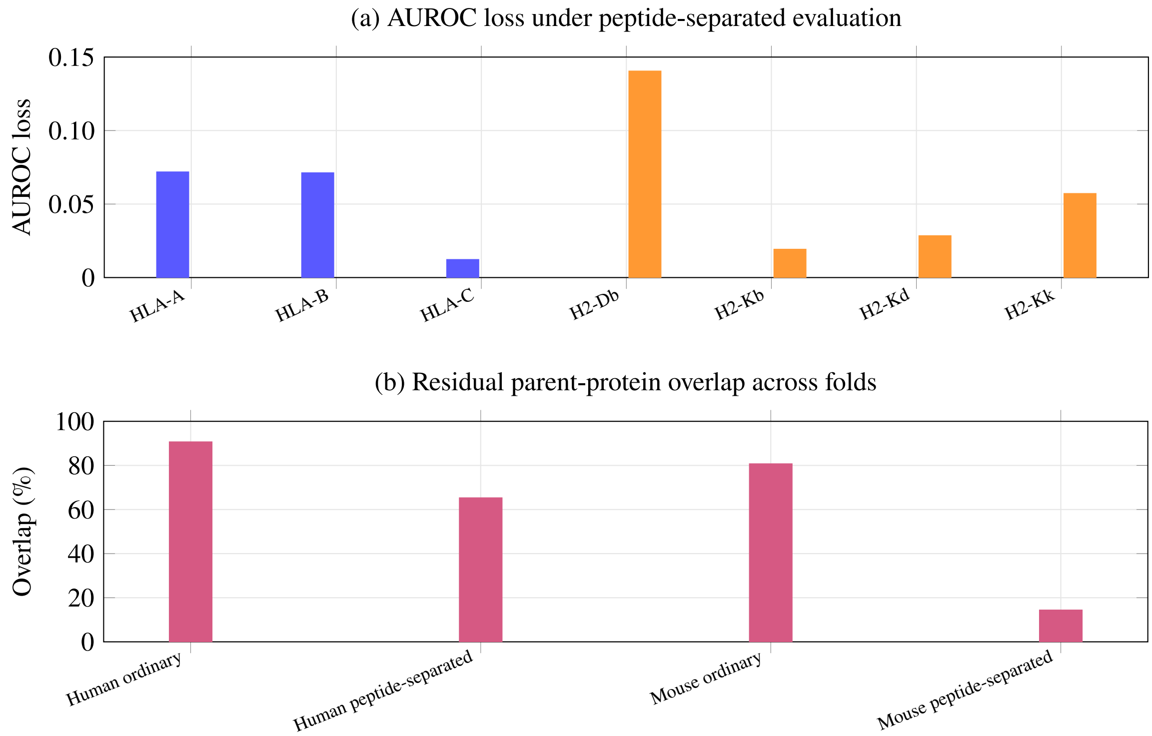

Figure A1: Generalization diagnostics using the same pooled evaluation rows. (a) Mean AUROC loss, ordinary minus peptide-separated cross-validation, by HLA gene group or H2 restriction; positive values indicate lower performance when held-out peptide sequences are absent from fitting. (b) Mean fraction of unique held-out parent proteins also observed during fitting; nonzero values quantify residual protein overlap.

B. Detailed Model Comparisons When Held-out Peptides Are Absent from All Fitting Combinations

Table A2: Selected paired AUROC comparisons between models evaluated on the same tissue–MHC combinations and peptide-separated folds. Peptide pairs connected directly or indirectly by a shared sequence are assigned to the same fold. Differences are calculated as the first model minus the second; intervals are based on 10,000 resamples of complete tissue–MHC combinations, and q values are adjusted within each species–metric family using the Benjamini–Hochberg procedure.

Comparison	Mean	Median	Higher/tied/ lower	95% CI	Adjusted q
Human					
TissuePMHC − shared peptide encoder with combination-specific outputs	+0.0091	+0.0095	118/0/39	[0.0067, 0.0116]	2.27×10^{-11}
TissuePMHC − unguided two-view neural network	+0.0097	+0.0081	122/0/35	[0.0076, 0.0119]	1.36×10^{-14}
TissuePMHC − guided two-view neural network	+0.0010	-0.0008	76/1/80	[-0.0008, 0.0029]	0.442
TissuePMHC − BLOSUM62 random forest	+0.0440	–	149/0/8	[0.0392, 0.0489]	3.43×10^{-26}
TissuePMHC − one-hot logistic regression	+0.0525	–	157/0/0	[0.0479, 0.0571]	8.14×10^{-27}
Guided two-view neural network − unguided two-view neural network	+0.0087	–	144/0/13	[0.0074, 0.0100]	3.02×10^{-23}
Guided two-view neural network − shared peptide encoder with combination-specific outputs	+0.0081	–	123/0/34	[0.0058, 0.0103]	3.30×10^{-11}

Table A2: Selected paired AUROC comparisons under peptide-separated cross-validation (continued).

Comparison	Mean	Median	Higher/tied/ lower	95% CI	Adjusted q
Mouse					
Dual branch with auxiliary supervision − dual branch without auxiliary supervision	+0.0034	+0.0031	21/0/3	[0.0023, 0.0047]	3.93×10^{-6}

Table A3: Stability across independent training runs and the benefit of averaging their predictions under peptide-separated cross-validation. Peptide-linked groups are kept intact when the folds are formed. Gain is the AUROC obtained after averaging predictions minus the mean AUROC of the separate runs.

Species	Model	Mean run	Across-run standard deviation	Averaged predictions	Gain
Human	One-hot logistic regression	0.7127	0.0000	0.7127	0.0000
Human	BLOSUM62 random forest	0.7193	0.0006	0.7212	+0.0019
Human	Shared peptide encoder with combination-specific outputs	0.7313	0.0043	0.7561	+0.0248
Human	Unguided two-view neural network	0.7394	0.0043	0.7555	+0.0161
Human	Guided two-view neural network	0.7503	0.0038	0.7642	+0.0139
Human	TissuePMHC	0.7505	0.0007	0.7652	+0.0147
Mouse	One-hot logistic regression	0.7335	0.0000	0.7335	0.0000
Mouse	BLOSUM62 random forest	0.7491	0.0012	0.7507	+0.0016
Mouse	Shared peptide encoder with combination-specific outputs	0.7327	0.0031	0.7538	+0.0211
Mouse	Dual branch without auxiliary supervision	0.7381	0.0031	0.7556	+0.0175
Mouse	Dual branch with auxiliary supervision	0.7422	0.0028	0.7590	+0.0167

C. Details of Tissue-blind General Presentation Controls

Table A4: Sensitivity analysis comparing the NetMHCpan binding-affinity score with its eluted-ligand score. Both rankings are oriented so that larger scores indicate stronger predicted propensity.

Species	Evaluation	Binding AUROC	Presentation AUROC	Difference
Human	Saved test	0.6577	0.6833	+0.0256
Human	Complete training pool	0.6559	0.6823	+0.0264
Mouse	Saved test	0.6847	0.6893	+0.0046
Mouse	Complete training pool	0.6778	0.6802	+0.0024

Table A5: Combining tissue-independent peptide–MHC scores with the human tissue-conditioned model score. The combination rule is learned on other folds, and all external scores were generated before this analysis.

Species	Protocol	External control	External only	Combined
Human	saved internal test set	MHCflurry predicted presentation	0.6851	0.8453
Human	saved internal test set	NetMHCpan eluted-ligand score	0.6833	0.8447
Human	Peptide-separated cross-validation	MHCflurry predicted presentation	0.6867	0.7693
Human	Peptide-separated cross-validation	NetMHCpan eluted-ligand score	0.6740	0.7651

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